

Exploring AAB patterns in degree morphology: A DM–Nanosyntax comparison

Jonathan David Bobaljik^{1*†} and Pavel Caha^{2*†}

^{1*}Department of Linguistics, Harvard University, Boylston Hall 3rd fl,
Cambridge, 02138, MA, USA.

²Department of Czech Language, Masaryk University, Arne Nováka 1,
Brno, 60200, Czechia.

*Corresponding author(s). E-mail(s): bobaljik@fas.harvard.edu;
pavel.caha@phil.muni.cz;

†These authors contributed equally to this work.

Abstract

The paper investigates stem morphology in triplets like *bad–worse–to worsen* (positive – comparative – deadjectival degree-achievement verb). It focuses on Czech, Ukrainian, Bulgarian and Turkish, and complements a previous typological study by Bobaljik (2012), which focused on root suppletion within this triplet of forms. The study replicates the main result of Bobaljik (2012) for stems, which is that ABA patterns are virtually unattested, confirming the idea that there is a containment relation among the three forms. The study also finds a relative wealth of AAB patterns, which are absent in root suppletion. The empirical record further suggests that AAB patterns correlate with a particular type of marking in the comparative, where the comparative marker is both a separate word and a pre-marker (it precedes the adjective). We argue that the theories of Distributed Morphology and Nanosyntax each naturally rely on a different facet of this data. In DM, the correlation between AAB patterns and the type of comparative marking is best explained using the notion of an independent word (where wordhood defines a locality domain, as in Bobaljik 2012). On the other hand, Nanosyntax naturally relies on the complementary characteristic, namely the pre-marking nature of the comparative (relying on the idea that pre-marking leads to the opening of a separate derivational workspace, see De Clercq and Vanden Wyngaerd 2018). We thus identify an empirical domain where future research may corroborate one or the other approach by identifying data that would differentiate between the two properties as the relevant correlate for the newly discovered AAB patterns.

Keywords: *ABA, Augment, Deadjectival verbs, Distributed Morphology, Nanosyntax, Suppletion

1 Introduction

Bobaljik (2012) investigates root suppletion in adjectival gradation. One finding, extending earlier work (see Ultan 1972), is the Comparative-Superlative-Generalization in (1) illustrated in the table in (2).

- (1) *The Comparative-Superlative Generalization* (CSG, Bobaljik 2012, 2):
If the comparative degree of an adjective is suppletive, then the superlative is also suppletive (i.e., with respect to the positive).

(2)

	POSITIVE	COMPARATIVE	SUPERLATIVE	GLOSS
a. English:	good	bett -er	be -st	
b. English:	bad	worse	wor -st	
c. Danish:	god	bed -re	bed -st	‘good’
d. Czech:	špatn-ý	hor -ší	nej- hor -ší	‘bad’
e. Estonian:	hea	pare -m	par -im	‘good’
f. Kildin Saami:	šig’	per ’-am	per ’-mus	‘good’
g. Basque:	asko	gehi -ago	gehi -en	‘many’
h. late Latin	bon-us	mel -ior	opt -imus	‘good’

Patterns such as **good* – *bett*-er – *good*-est, the so-called *ABA patterns, where the same root occurs in the positive and the superlative, but the comparative alone is suppletive, are virtually unattested in (synthetic) constructions.¹ Bobaljik (2012) also notes that a parallel generalization seemingly holds of root suppletion in triplets involving the positive degree adjective (*wide*), its comparative (*wider*) and a corresponding degree achievement verb (*widen*). A representative selection of triplets where root suppletion is attested is provided in Table (3).

(3)

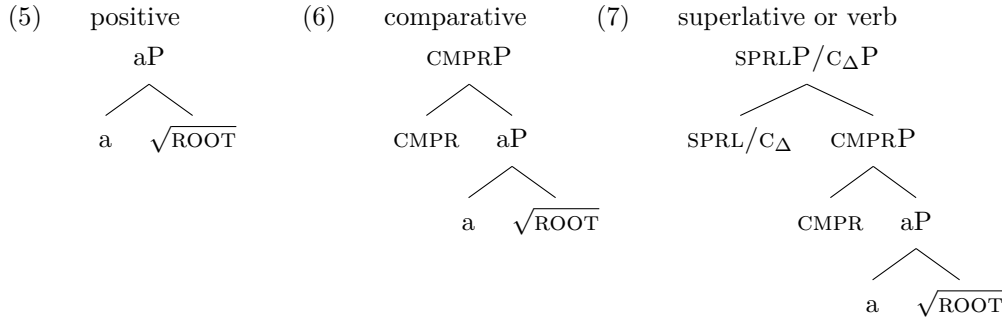
	POSITIVE	COMPARATIVE	VERB	GLOSS
English	good	bett -er	bett -er	
English	bad	worse	wors -en	
German	gut	bess -er	ver- bess -er-n	‘good’
Russian	plox-oj	xuž -e	u- xud -š-ať	‘bad’
Finnish	hyvä	pare -mpi	para -ntaa	‘good’
Georgian	cud-i	u- ar -es-i	a-u- ar -es-ebs	‘bad’
late Latin	bon-us	mel -ior	mel -iōr-o	‘good’

¹See Bobaljik (2012) for various qualifications. For example, the generalization holds of ‘relative’ superlatives – those meaning ‘more A than all others’, but not of ‘absolute’ superlatives or elatives – those just meaning A to the highest degree.

The data in the table show that when the comparative is based on a suppletive root, so is the change-of-state verb; triplets such as the hypothetical **good-better-gooden* are rather rare. Bobaljik (2012) expresses this state of affairs by proposing the generalization in (4).

- (4) *The Comparative–Change of State Generalization* (CΔG, Bobaljik 2012, 6):
If the comparative degree of an adjective is suppletive, then the corresponding change-of-state verb is also suppletive (i.e., with respect to the positive adjective).

Bobaljik (2012, 171) explains both generalizations by proposing (syntactic) structures (from which corresponding morphological structures are derived) of growing complexity like the ones in (5)–(7). Here CMRP stands for the comparative, SPRL for superlative and C_Δ carries a change-of-state meaning. We have added *a*, a categorizing functional head found by hypothesis in all adjectives if roots are ‘category-neutral’ (Marantz 1997; Borer 2005 a.o.).



The core idea is that the comparative is based on the positive by adding the CMRP head, compare (5) and (6). The superlative and change-of-state verb are in turn built on top of the comparative by the addition of the superlative or change-of-state head, C_Δ, as in (7).

The CSG and CΔG generalizations (1) and (4) follow from this under the idea that root suppletion in the comparative is triggered by the CMRP head. Since this head is also found in the superlative and the verb, there is no way for the verb to not be suppletive if the comparative is.² A further assumption in Bobaljik (2012) is that suppletion only applies if the heads in structures like (5)–(7) are collected together into a complex head, via head movement or some alternative. This aspect of the theory will become relevant in Sections 4-5 below.

The CSG and CΔG correctly exclude ABA patterns. Alongside the well attested AAA, ABB, and rare ABC patterns (as in the Latin example in (2)), they admit, at

²These containment structures offer a plausible explanation of why the CSG and CΔG are limited to relative superlatives and deadjectival degree achievements, since it is the presence of the comparative head that triggers the suppletion, thus it is only those structures with comparative semantics (degree achievements) that are expected to be subject to the generalizations. Deadjectival verbs with non-comparative semantics, such as ‘belittle’, are not expected to be suppletive even in languages with a suppletive comparative of ‘little’. The relevant semantic distinctions are not trivial; see Kennedy and Levin (2008), Bobaljik (2012, Ch6) and references there for some discussion.

least in theory, of AAB patterns, where the positive and comparative share one form of the root (AA), while the corresponding superlative, or, for our interests, deadjectival verb has a suppletive root (B). Such patterns are not ruled out by the structures in (5)-(7), since the suppletive root in the verb (i.e., the B root) could be conditioned by the head C_{Δ} found only in verbs (or SPRL in superlatives). While AAB patterns do occur in some domains where *ABA patterns are absent (suggesting that they should be treated separately), Bobaljik (2012) noted that the AAB pattern is unattested in superlatives (and offered an account). The matter was left open for verbs where the empirical (non-)existence of such patterns seems to be impossible to establish conclusively for roots (see Bobaljik 2012, 174). The reasons for this have to do with both the (non)productivity of category-changing morphology and near polysemy in lexical (verbal) roots. To see the issue, consider the table in (8).

(8)

POSITIVE	COMPARATIVE	VERB	HYPOTHETICAL
<i>small</i>	<i>small-er</i>	???	shrink, diminish, lessen
<i>tall</i>	<i>tall-er</i>	???	grow

The table shows that an adjective such as *small* does not have a corresponding change-of-state verb based on the same root (**to small(en)*). However, it is not clear whether this cell in the paradigm is simply empty (change-of-state verbs are unproductive) or whether one of the semantically fitting verbs (listed in the final column) should be slotted into the relevant cell as an instance of suppletion. Therefore, the cell is marked with question marks as its status is in principle unclear. The same situation arises for *tall* and other adjectives.

In the current paper, we shift our attention away from root allomorphy to allomorphy of ‘stem markers (SM)’ (also known as ‘augments’) that is, suffixes that combine with roots to form adjectival stems. Our focus will be suffixes like the *-k* in Czech *slad-k-ý* ‘sweet-SM-INFL’ and cognates in Ukrainian and Bulgarian, as well as an apparently similar suffix (also *-k*) in Turkish. We assume for the purposes of this paper that such stem markers are exponents of little *a* in the structures in (5)-(7).³ From a Nanosyntax or DM perspective, allomorphy of stem markers should be subject to the same restrictions as allomorphy of roots, at least if nothing else is added to the theory (see Zoppi 2017 and McFadden 2018, and more generally also Gouskova and Bobaljik 2024).

This will allow us to avoid the indeterminacy that arises in suppletion. Our first empirical results are that we can show that the pattern of stem markers do indeed obey the same restriction as roots, i.e., their allomorphy complies with the $C_{\Delta}G$ generalization, but moreover, that AAB patterns do arise in adjective – comparative – verb triples.

But this result in turn opens a new set of questions. While AAB patterns are attested, their distribution vis-à-vis ABB patterns is not random. It turns out that our small language sample splits into two groups.

³We recognize that there are other views about stem markers in general and about Slavic stem augments in particular, but as far as we can tell, nothing of substance hinges on this simplification.

In Czech and Ukrainian, the distribution of stem-marker allomorphy replicates what is already described for root suppletion. Namely, some roots that combine with a stem marker in the positive (root+SM) lose that marker in the comparative (root-CMPR). When they do so, they also lack the marker in the change-of-state verb as well as in the superlative (root-VERB/SPRL). Some comparatives retain the stem marker (root-SM-CMPR), and the behaviour in the comparative is inherited by the change-of-state verbs and the superlatives (root-SM-VERB/SPRL). With a handful of exceptions in one type of inchoative, the attested patterns are thus AAA and ABB.

By contrast, in Bulgarian and Turkish, the only pattern (we will argue) with change-of-state verbs is AAB. We observe that the prevalence of AABs in these languages correlates with the fact that comparatives in Bulgarian and Turkish are not expressed by suffixes, as in Czech and Ukrainian, but by markers that precede the root and show hallmarks of independent words. We formulate this as the AAB conjecture (9).

- (9) The AAB conjecture
Systematic AAB patterns arise in languages with analytic, pre-marking comparatives and suffixal change-of-state verbs.

We argue that the theoretical assumptions of both Nanosyntax and DM as used to describe root suppletion in fact predict this difference in the distribution of AAB patterns, and, moreover, derive it from the different morphosyntax of comparatives in Bulgarian and Turkish.

In keeping with the theme of this special collection, we show that while Nanosyntax and DM both derive the cross-linguistic differences that we observe, they do so in different ways, and specifically in ways that potentially lead to differing empirical predictions that could be profitably explored in future work.

The paper is organized as follows. In Section 2, we introduce Czech and Ukrainian adjectives with stem markers and show that their distribution exhibits robust ABB patterns. This places the stem markers firmly within the area covered by the CΔG (4). In Section 3, we turn to Bulgarian and show that in this language, the stem markers that are cognate with those in Czech and Ukrainian exhibit a robust AAB pattern. We hypothesize that this correlates with another feature of Bulgarian, namely that the comparative precedes the adjective and corresponds to an independent word. We also introduce Turkish as a language where the same correlation between AAB patterns and the morphological status of the comparative is observed. In Sections 4 and 5, we present an account of these facts in Nanosyntax and Distributed Morphology respectively. Section 6 zooms in on an unexpected case of root suppletion in Bulgarian and shows how the data can be made compatible with each of the frameworks. Section 7 concludes by presenting a discussion of the commonalities and differences in the theoretical approaches. Our main point is that each of the frameworks picks on a different facet of the data (the prefixal status or prosodic independence of the comparative). These happen to coincide in the languages we look at (i.e., Bulgarian and Turkish), which raises the possibility that in languages where they do not coincide, one of them will prove to be the decisive force for AAB patterns. We thus identify an

area where future research may provide some empirical evidence that would allow us to distinguish the two approaches on empirical grounds.

2 Czech and Ukrainian causatives: ABB not AAB

This section examines patterns of stem allomorphy in Czech and Ukrainian. In Section 2.1, we show that Czech has at least two stem markers that are found in the positive degree but may be missing in the comparative (see Vanden Wyngaerd et al. 2025). We demonstrate that whatever happens to the stem marker in the comparative, the same thing happens to it in the causative change-of-state verb. These stem markers thus replicate the behavior of roots: when the stem marker has an irregular form in the comparative, it is the comparative form that is inherited by the verb. Section 2.2 illustrates the same thing for Ukrainian based on the work by Vyshnevskya (2025a).

2.1 ABBs in Czech

In this section, we discuss Czech causative verbs formed from adjectives with complex stems. We show that the stem marker is dropped in the comparative of some adjectives, but retained with others. Crucially, no matter what behavior the adjective shows in the comparative (i.e., whether it drops the stem marker or not), the same behavior is found in the verb. This finding replicates CΔG (4) in the domain of stems.

Table (10) illustrates this pattern for the stem marker *-k*.⁴

(10) Czech adjectives with the stem marker *k*

gloss	adjective	comparative	causative
clean	čist- -ý	čist- š-í	čist- i-t
tough	tuh- -ý	tuž- š-í	tuž- i-t
sweet	slad- k -ý	slad- š-í	slad- i-t
tame	krot- k -ý	krot- š-í	krot- i-t
bitter	trp- k -ý	trp-č-š-í	trp-č-i-t
thin	ten- k -ý	ten-č-š-í	ten-č-i-t

To see the pattern in the relevant context, consider first the adjectives in the upper section of Table (10). These correspond to a bare root (followed by the obligatory agreement markers). With these adjectives, we can observe the regular comparative marker *-š* and the causative marker *-i* (which is further followed by the infinitive marker *-t*). Note that both the comparative *-š* and the causative *-i* trigger consonant mutations; e.g., the root final *h* of the root *tuh* ‘tough’ on the second row changes to *ž*.

The remaining horizontal sections of the table show four adjectives that are morphologically complex in the positive degree, featuring the stem-marker *-k*. In the

⁴The stem marker *-k* gets palatalised to *-č* both in the comparative and in the change-of-state verb; this is a regular process in Czech. Another regular process is that the sequence *čš* in the comparative is simplified to *č*, so orthographically, we get *trpč-í* ‘more bitter’ and *tenč-í* ‘thinner’ in Czech orthography. In our tables, we will always indicate the morphological analysis, and write *č-š*, but ask the reader to keep in mind that in the written language, only *-č* is found in these cases.

middle section, we observe two adjectives where the stem marker is dropped in the comparative, and the comparative *-š* attaches directly to the root. For such adjectives, the causative *-i* also attaches to the bare root. The distribution of the stem marker in these adjectives (*k-ø-ø*) resembles the ABB pattern of root suppletion (*bad-worse-wors-en*).⁵

In the lowermost section, we see two adjectives where the stem marker *-k* is preserved in the comparative. With these adjectives, the stem marker is preserved also in the causative verb (AAA pattern).

It is not the case that the adjectives that preserve the stem marker are morphologically simplex. This can be shown by diminutive adjectives, see (11).

- (11) Diminutive adjectives
- a. *slad*¹-ouč-*k*-ý
sweet-DIM-SM-AGR
'very sweet'
 - b. *ten*²-ouč-*k*-ý
thin-DIM-SM-AGR
'very thin'

The data point in (11a) shows the adjective 'sweet.' We saw in Table (10) that the stem marker is a separate morpheme (since it drops in the comparative). (11a) shows the same thing: we can see that *-k* is a separate morpheme because it is separated from the root by the diminutive.⁶

Particularly relevant is the adjective *ten-k-ý* 'thin' in (11b). This adjective preserves the stem marker in all forms in Table (10) (see the bottom row), yet still places the diminutive between the root and the stem marker. This shows that also adjectives which preserve the stem marker in the comparative and in the verb are morphologically complex.

The table below provides a complete overview of all adjectives that have the stem marker *-k* and form a causative. The correlation between the fate of the stem marker in the comparative and in the corresponding verb is perfect, as the shading brings out. Such a pattern clearly falls under the CΔG: just like we have no ABA in root suppletion cases, i.e., no **bad-worse-badden*, there is no ABA pattern of stem marker distribution; i.e., no *k-ø-k* pattern. When there is a zero allomorph in the comparative, there is a zero allomorph in the verb.⁷

⁵See Caha et al. (2023) for more on Czech deadjectival verbs and the role theme vowels play. Another point worth noting is that the deadjectival verbs often appear with perfectivizing prefixes. We abstract away from these prefixes here, though some verbs sound much more natural with a prefix included. We will note this once again when we discuss Bulgarian in Section 3.1.

⁶Note that the stem marker is obligatory following the diminutive, even for roots that do not otherwise take the stem marker, e.g., (Czech): 'small' = *mal-ý* → *mal-in-k-ý* (**mal-in-ý*), 'weak' = *slab-ý* → *slab-ouč-k-ý* (**slab-ouč-ý*) etc. That is, when root-adjacent, the appearance of the stem marker is tied idiosyncratically to particular roots, but in the diminutive its appearance is root-independent.

⁷In the table, roots always shorten in the comparative, e.g., *blíz-k-ý* 'close' has a long vowel *í* in the positive and in the causative, but a short vowel in the comparative. In Scheer (2003), this is treated as a templatic effect that targets all Czech comparatives in *-š-í*, and we therefore treat these examples as involving the same lexical item, giving rise to an AAA pattern of root distribution.

It is important to make it explicit that choices like these (i.e., whether two forms are suppletive or just distinct phonological realizations of the same root) have consequences for the inventory of patterns. Bobaljik (2012, 140-2) shows that there are patterns which would have to be treated as **ABA* without assuming any phonological processes, but can be understood as AAA patterns if one admits the existence of a phonological

(12) Czech ABBs/AAAs: the causative patterns with the comparative

gloss	adjective	comparative	causative
close	blíz- k -ý	blíž- š-í	blíž- i-t
smooth	hlad- k -ý	hlad- š-í	hlad- i-t
short	krát- k -ý	krat- š-í	krát- i-t
low	níz- k -ý	niž- š-í	s-níž- i-t
narrow	úz- k -ý	už- š-í	úz- i-t
tame	krot- k -ý	krot- š-í	krot- i-t
sweet	slad- k -ý	slad- š-í	slad- i-t
thin	říd- k -ý	řid- š-í	řed- i-t
heavy	těž- k -ý	těž- š-í	tíž- i-t
light	leh- k -ý	leh-č-š-í	leh-č-i-t
bitter	trp- k -ý	trp-č-š-í	trp-č-i-t
soft	měk- k -ý	měk-č-š-í	měk-č-i-t
thin	ten- k -ý	ten-č-š-í	ten-č-i-t
fragile	křeh- k -ý	křeh-č-š-í	křeh-č-i-t
wet	vlh- k -ý	vlh-č-š-í	vlh-č-i-t

The same pattern surfaces with adjectives that use a different stem marker, namely *-ok*. This is a small class of adjectives, yet we find the same abstract pattern. The full list of such adjectives is provided in (13).

(13) Czech: a different stem marker, the same two patterns

glos	POS	CMPR	CAUS
deep	hlub- ok -ý	hlub- š-í	hloub- i-t
wide	šir- ok -ý	šir- š-í	šír- i-t
tall	vys- ok -ý	vyš- š-í	výš- i-t
wild	div- ok -ý	div- oč -ejš-í	div- oč -i-t

(13) shows exactly the same pattern as (12): when the stem marker is absent in the comparative, it is also absent in the causative. The single adjective that preserves this marker in the comparative also preserves it in the causative.⁸

With this piece of data, we conclude the discussion of the Czech causatives. The point was to show that there is a clear and non-trivial relationship of the stem shapes

rule that applies in the comparative and changes an underlying A to a slightly different surface form, B. This is the case also with roots like *blíz/bliz*, which have a short vowel only in the comparative.

The middle section of the table lists two roots where the causative has an irregular shape that seems difficult to explain assuming regular phonological processes targeting all causatives. These cases could thus constitute an AAB pattern of root suppletion. However, different theories treat cases of weak suppletion like this in different ways, and some would analyze such examples as involving the same root. We therefore do not draw any conclusions on the basis of this example, though it is interesting in a larger perspective.

⁸The comparative has a different allomorph with the last adjective (*-ejš*). This does not affect our point, see Caha et al. (2019); Vanden Wyngaerd et al. (2025) for discussion.

in the comparative and in the change-of-state verb, which can be summarized as follows:

- (14) a. If the stem marker is preserved in the comparative, it is also preserved in the causative change-of-state verb
 b. If the stem marker is dropped in the comparative, it is also dropped in the causative change-of-state verb

This establishes a parallel between the behavior of roots under suppletion (as discussed in Bobaljik 2012) and the behavior of stems. This parallel is expected under the view that the change-of-state verb is based on the comparative and that the little *a* head reflects the same structural environment as the root.

Before we leave this section, we would like to point out that as far as we are aware, the distinction between the verbs in the AAA pattern and the ABB pattern is purely morphological. Thus, the comparative has the same syntax and meaning regardless of whether it includes the stem marker or not. The verbs of both types appear to be degree achievements (see Bobaljik 2012, 176-187 and references therein) and neither type entails that the positive is true of the result state. That is, both morphological types appear to embed the comparative meaning, regardless of whether they contain an augment or not. We show this in (15) and (16). In such sentences, the verb has to be translated using a comparative rather than the positive degree of the adjective.

- (15) Emma měla vždycky **krát-k-é** vlasy, a teď si je ještě z-**krát -i-la**
 Emma had always short hair and now REFL it still short-CAUS-PAST
 ‘Emma has always had short hair, and now she made it even shorter.’
 (16) Výrobci se podařilo z-**ten-č-i-t** už tak **ten-k-é** komponenty.
 producer REFL managed thin-SM-CAUS-INF already thin-SM-AGR components
 ‘The producer managed to make the components, which were already thin, thinner.’

We shall later show that such a comparative meaning is also found in languages that show an AAB pattern. The presence/absence of the augment thus appears to be purely morphological, with no consequences for the meaning or syntax of the relevant comparatives or verbs.⁹

⁹We are leaving aside Czech inchoatives, derived by the verbalizer *-nou*, which provide an even richer set of patterns, as illustrated in (i). On the top of the table, we see a familiar ABB pattern, and below it a likewise familiar AAA pattern. Curiously, there are two adjectival roots in Czech exhibiting an AAB pattern and one root with an apparent ABA pattern.

(i) Inchoative verbs in Czech

gloss	adjective	comparative	inchoative
tame	krot- k-ý	krot- š-í	krot - nou-t
sweet	slad- k-ý	slad- š-í	slád - nou-t
light	leh- k-ý	leh- č-š-í	leh- k-nou-t
bitter	trp- k-ý	trp- č-š-í	trp- k-nou-t
fragile	křeh- k-ý	křeh- č-š-í	křeh - nou-t
wet	vlh- k-ý	vlh- č-š-í	vlh - nou-t
heavy	těž- k-ý	těž- š-í	těž- k-nou-t

2.2 Ukrainian

Ukrainian (Vyshnevskya 2025a) exhibits ABB patterns (alongside AAA patterns) of exactly the same type as Czech. Table (17) shows selected triplets of the positive, comparative and the corresponding inchoative change-of-state verb.¹⁰

(17) Ukrainian triplets positive-comparative-inchoative (Vyshnevskya 2025a)

gloss	positive	comparative	inchoative
merry	vesel- -yj	vesel- -iš-yj	vesel- -iš-a-ty
young	molod- -yj	molod- -š-yj	molod- -š-a-ty
short	korot- k -yj	korot- -š-yj	korot- -š-a-ty
thin	ton- k -yj	ton- -š-yj	ton- -š-a-ty
light, easy	leh- k -yj	leh- -š-yj	leh- -š-a-ty
weak	slab- k -yj	slab- -š-yj	slab- -š-a-ty
clear	čit- k -yj	čit- k -iš-yj	čit- k -iš-a-ty
fit	strun- k -yj	strun- k -iš-yj	strun- k -iš-a-ty
bouncy	pruh- k -yj	pruh- k -iš-yj	pruh- k -iš-a-ty
weak	slab- k -yj	slab- k -iš-yj	slab- k -iš-a-ty

The topmost section of the table shows two morphologically simplex adjectives where in the positive, the adjectival agreement *-yj* attaches directly to the root. The comparative has two allomorphs, *-š* and *-iš*. The final column shows that inchoative change-of-state verbs are formed by attaching the verbal theme marker *-a* on top of the full comparative (minus agreement), preserving the comparative *-š/-iš* allomorphy.

The two remaining sections of the table show adjectives that exhibit the stem marker *-k* in the positive. In the comparative, they fall into two classes. The first class (in the middle section of the table) loses the stem marker in the comparative, while the second class (in the bottom section) retains it. The examples are chosen to represent phonologically similar roots (ending in *Vt*, *Vn*, *Vh* and *Vb*) going top down, which makes it unlikely that these patterns arise due to phonological deletion.¹¹

The final column shows the corresponding change-of-state verb. We can see that the situation is the same as in Czech: whenever the stem marker is absent in the comparative, it is also absent in the verb. When it is present in the comparative, it is present in the verb as well.

We leave these cases aside. As for describing AAB in Czech, this would pose no problem, but we rather focus in this paper on the fact that AAB patterns are much more prevalent in languages with analytic/pre-adjectival comparative marking, while the ABB pattern, robustly attested in Czech, is absent there. As for the suspected *ABA pattern, it was noted already in Bobaljik (2012, 6.4.3-4) that Slavic inchoatives are a subdomain with other irregularities including apparent violations of the CΔG (*ABA) for root suppletion. For some thoughts on why sporadic apparent counter-examples show up in various subdomains where *ABA generally holds, see also Bobaljik and Sauerland (2018), Blix (2022), Caha (2025), Van Craenenbroeck (2025), Starke and Dalen (2026).

¹⁰We do not discuss causative verbs in Ukrainian since Vyshnevskya (2025a, sec. 8.1) sets them aside as they are less productive than inchoatives.

¹¹Note that the adjective ‘weak’ shows variably either an ABB or an AAA pattern. This strengthens the point that the distribution of arguments is unrelated to phonology or meaning, since, obviously, with such adjectives, the same phonology and meaning is compatible with both patterns.

The grammatical relevance of the pattern is underscored by the fact that Ukrainian (like Czech) also shows additional stem markers that follow the same pattern, see (18) for illustration.

- (18) Adjectives with the stem marker *-ok* (Vyshnevskya 2025a)

gloss	positive	comparative	inchoative
wide	šyr- ok -yj	šyr- š -yj	šyr- š - a -ty
cruel	žorst- ok -yj	žorst- ok - iš -yj	žorst- ok - iš - a -ty

It is further worth pointing out that Ukrainian inchoatives differ from those in Czech and maintain the comparative *-š/-iš*. This state of affairs further strengthens the case for an analysis where the absence of the stem marker in the verb is linked to its fate in the comparative: Ukrainian brings this out clearly, since the whole comparative is literally contained in the verb. Once we have an approach that explains the distribution of *-k/-ok* in the comparatives (using the same ingredients as already proposed in Bobaljik 2012 for roots), it can be directly extended the change-of-state verb. The same approach can then be further extended to Czech and similar cases, where the only extra proviso must be made with respect to the fate of the comparative marker in the verb (realized as $\emptyset$ when embedded under C_{Δ}).¹²

2.3 ABBs: an intermediate summary

As in other Slavic languages, some Czech and Ukrainian adjectival roots are lexically specified such that they combine with a stem-marker *-k/-ok* in the positive. This class

¹²Vyshnevskya (2025a) points out that Ukrainian has two kinds of inchoatives, which she calls inchoatives 1 and 2. Vyshnevskya's inchoative 1 verbs are discussed in the main text and they uniformly exhibit either an AAA or an ABB pattern. Inchoative 2 verbs are less productive and they use the stem marker *-i*. As shown in Table (i), this marker is never preceded by *-k*, and it therefore, gives rise to ABB and AAB patterns. Once again, the AAB pattern is not problematic for the structures proposed here, in fact, its existence is what we want to demonstrate. However, what we find striking is its scarcity in languages like Czech and Ukrainian (two examples in Czech, one in Ukrainian) and its dominance in Bulgarian and Turkish, which we shall turn to shortly.

- (i) Inchoative 2 verbs in Ukrainian (Vyshnevskya 2025a)

gloss	positive	comparative	inchoative 2
short	korot- k -yj	korot- š -yj	korot -i -ty
sweet	solod- k -yj	solod- š -yj	solod -i -ty
weak	slab- k -yj	slab- š -yj	slab -i -ty
shallow	mil- k -yj	mil- k - iš -yj	mil -i -ty

In addition to the rare AAB pattern in stem formation, Ukrainian inchoative 2 verbs also exhibit ABA patterns of root suppletion, as in (ii).

- (ii) Ukrainian (Vyshnevskya 2025a, 172)

gloss	adjective	comparative	inchoative 1	inchoative 2
bad	pohan -yj	hir -š-yj	hir -š-a-ty	pohan -i-ty

We shall not discuss these cases here, see the discussion in footnote 9. In addition to Vyshnevskya's work, see also Bobaljik (2012, 204) for more on Ukrainian ABAs. See Dees and VanDyne (ress) for a discussion of ABAs of a similar kind in Polish.

is further divided into a group that retains the stem-marker in the comparative, and a group that loses it. What we have shown here is that the behaviour in the comparative is predictive of the behaviour not only in the superlative, as expected on the grounds of CSG, but also in all causative change-of-state verb in Czech and all *-a* inchoatives in Ukrainian, which only exhibit AAA and ABB patterns.¹³

We interpret this as an instance of the CΔG generalization proposed by Bobaljik (2012). What lies at the heart of the explanation is the hypothesis that deadjectival verbs are uniformly formed on top of the comparative structure, as in (5)-(7). Once such structures are adopted, the allomorphy for little *a* carries over from the comparative to the verb, since in both forms, little *a* is found in the same configuration, namely in between the root and the comparative. This abstract containment relation is manifest in Ukrainian, where the comparative is literally contained in the relevant verbs.

3 Bulgarian and Turkish: AAB not ABB

We now turn to Bulgarian (Section 3.1) and Turkish (3.2). The patterns involving stem-markers in these languages differ strikingly from Czech and Ukrainian. On the one hand these languages are simpler in that they show no irregularity in the comparative: stem-markers are always retained in the comparative. On the other hand, they are always lost in the change-of-state verbs. Rather than the AAA and ABB patterns we have seen in Czech and Ukrainian, what we find in Turkish and Bulgarian is exclusively AAB patterns. After presenting the empirical evidence, we argue that the difference in patterns is correlated with a difference in how the comparative is marked: in Czech and Ukrainian the comparative is a suffix, while in Bulgarian and Turkish, the comparative is marked to the left of the adjective and by a marker that shows the properties of an independent word (like English *more*). This is the correlation that we formulated in the Introduction as the AAB conjecture (9).

(19) The AAB conjecture

Systematic AAB patterns arise in languages with analytic, pre-marking comparatives and suffixal change-of-state verbs.

The intuition behind this conjecture is that prefixal and analytic marking is regular, and only suffixal marking leads to ‘irregularities.’ If all marking was prefixal and analytic, we would only ever get AAA patterns. Thus, in languages like Bulgarian and Turkish, with analytic, prefixal comparatives, we will always get sequences starting with AA. But since their change-of-state verbs are suffixal, AAB patterns may and do arise.

It is a priori unclear which factor (i.e., being pre-marking or being analytic – or both) is responsible for precluding irregularity. In Sections 4 and 5, we describe theoretical proposals in Nanosyntax and DM that rely on one or the other property to derive the general shape of the facts. However, more empirical research than we can present here is needed to settle this issue on empirical grounds.

¹³Two instances of AAB and one of ABA are found in Czech inchoatives and one further instance AAB in Ukrainian unproductive inchoatives 2 in *-i*, as discussed in footnotes 9 and 12.

3.1 Bulgarian

In Bulgarian, deadjectival verbs are generally formed with the suffix *-i*, as in Table (20).

(20) Deadjectival verbs in Bulgarian

	POS	CAUS, 3.SG
clean	čist	(iz-)čist-i
warm	topəl	(s-)topl-i
short	kəs	s-kəs-i

As we noted for Czech and Ukrainian (see fn. 5), verbs across Slavic can generally combine with prefixes. For deadjectival verbs in Bulgarian, these prefixes are sometimes obligatory, as in the last row, and other times optional, as in the first two rows. Despite the fact that deadjectival verbs may require both a prefix and a suffix, we will consider the suffix to be the main verbalizer.¹⁴

We now turn to the behaviour of stem markers in Bulgarian. Recall that in Czech and Ukrainian, the distribution of the stem marker in the comparative was unpredictable, but was correlated across the comparative and the deadjectival verbs, yielding AAA (stem marker retained in both) or ABB (stem marker lost in both) patterns. By contrast, in Bulgarian, the distribution of the stem marker *-ək* is completely predictable in that it is always retained in the comparative and dropped in the causative, yielding an AAB pattern.

(21) Stem marker *-ək* shows an AAB pattern in Bulgarian

	POS	CMPR	CAUS (3.SG)
small	mal-ək	po-mal-ək	na-mal -i
short	krat-ək	po-krat-ək	sə-krat -i
smooth	glad-ək	po-glad-ək	iz-glad -i
thin	tən-ək	po-tən-ək	iz-tən -i
sweet	slad-ək	po-slad-ək	po-slad -i
heavy	tež-ək	po-tež-ək	na-tež -i
close	bliz-ək	po-bliz-ək	na-bliž -i
low	nis-ək	po-nis-ək	po-niž -i
tame	krot-ək	po-krot-ək	u-krot -i
strong	krep-ək	po-krep-ək	pod-krep -i
thin (liquid)	rjad-ək	po-rjad-ək	raz-red -i

¹⁴In the literature (Svenonius 2004), prefixes have been likened to English particles. Similarly to particles, the prefixes may influence argument structure (as in *They slept the party *(off)*), and they also contribute an aspectual property referred to as perfectivity (similar to, but distinct from telicity). On this account, it is the suffix that lexicalizes the verbal functional projections, including the relevant change of state head that is responsible for turning adjectives into verbs (see, e.g. Svenonius 2004, 239, Caha et al. 2023, Vyshnevskaya 2025a), which is also why it is obligatory.

Despite the fact that the verbs lack the stem marker that is found in the comparative, they preserve the comparative semantics as illustrated in (22) (compare the Czech example in (16)).

- (22) a. Nikoga ne eksfolirate kožata okolo očite,
 never NEG exfoliate skin.DEF around eyes.DEF
 b. tova šte iz-**tən-i** i bez tova **tən-k-a-ta** koža
 that will PREF-thin-CAUS even without that thin-SM-AGR-DEF skin
 ‘Never exfoliate the skin around your eyes, that will make the already thin skin even thinner/*thin.’

Thus, the real difference between Bulgarian on the one hand and Czech/Ukrainian on the other is not the nature of the verb, it is the nature of the comparative. While in Czech/Ukrainian, the comparative always patterns with the verb, in Bulgarian, it always patterns with the positive. Table (23) illustrates this on the example of several cognate adjectives. The upper part includes adjectives with augments. Uniformly, these lack the stem marker in Czech and Ukrainian comparatives, but have it in Bulgarian. In the lower section of the table, we show that the Bulgarian comparative patterns with the positive adjective also with property-concept adjectives exhibiting root suppletion in other Slavic languages.¹⁵

- (23) Stems in Czech and Ukrainian vs. Bulgarian (positive and comparative)

GLOSS	CZECH		UKRAINIAN		BULGARIAN	
	POS	CMPR	POS	CMPR	POS	CMPR
‘short’	krát- k-ý	krat-š-í	korot- k-yj	korot-š- yj	krat-ək	po krat-ək
‘heavy’	těž- k-ý	těž-š-í	tjaž- k-yj	tjaž-č- yj	tež-ək	po tež-ək
‘sweet’	slad- k-ý	slad-š-í	solod- k-yj	solod-š- yj	slad-ək	po slad-ək
‘good’	dobr-ý	lep-š-í	dobr-yj	lip-š- yj	dobər	po dobər
‘bad’	špatn-ý	hor-š-í	pohan-yj	hir-š- yj	loš	po loš
‘big’	velk-ý	vět-š-í	velyk-yj	bilj-š- yj	goljam	po goljam

Since the adjectival comparative always patterns with the positive, Bulgarian only allows patterns starting with AA. In (21), we showed that for the stem marker ək, such AA patterns are always expanded into AAB patterns. For the stem marker -ok, both AAB and AAA patterns are attested:

- (24) Stem marker -ok shows an AAB/AAA pattern in Bulgarian

	POS	CMPR	CAUS (3.SG)	
wide	šir-ok	po-šir-ok	raz-šir	-i
high	vis-ok	po-vis-ok	po-viš	-i
deep	dəlb-ok	po-dəlb-ok	za-dəlb-oč-	i

¹⁵Bulgarian exhibits root suppletion with the quantifier *mnogo* ‘many,’ comparative *poveče* ‘more’, which we address in Section 6 below.

These patterns, taken together, show a strong preference for AAB patterns, which seems absolute for the stem marker *-ə̌k*. To test for potential AAA patterns with *-ə̌k*, we searched the Bulgarian corpus bgTenTen21 on Sketch Engine (Kilgariff et al. 2014) for verbs with a stem in *-i* following a base ending in a consonant + *č*, i.e., the (palatalized) stem marker. This procedure was supposed to return verbs like *na-mal-i* ‘make smaller’ but with the stem marker preserved, i.e., hypothetical forms like **na-mal-č-i*. The search turned up a couple of potential candidates for an AAA pattern but it appears that all such examples seem to have a different analysis, namely one where the hypothetical stem marker has been reanalyzed as a part of the root. A concrete example is provided in (25). The table shows the adjective ‘bitter’ in Bulgarian and Czech. In the first row, it suggests a segmentation of the Bulgarian pattern where, apparently, the stem marker *-ə̌k* (palatalized to *-č*) is found in all forms, including the causative. This would seem to be parallel to the Czech example in the second row.

(25)

	POS	CMPR	CAUS	LANGUAGE
bitter/poor (emot.)	gor-ə̌k	po- gor-ə̌k	v- gor-č -i	Bulgarian
bitter	hoř-k-ý	hoř-č-š-í	roz- hoř-č -i-t	Czech

However, there are reasons to think that the segmentation provided for the Bulgarian *gor-ə̌k* in the table is incorrect and that the (etymologically independent) stem marker here has been reanalyzed as a part of the root. One argument to this effect comes from diminutive adjectives. In general, for adjectives that take the stem-marker *ə̌k*, the diminutive *-ič* comes between the root and the stem-marker, as in (26) (just as in Czech (11)). This, recall, provides converging evidence that the true stem marker is an independent affix, and is lost in the verbs.

- (26) **krat-ə̌k** — **krat-ič-ə̌k**
 short-SM — short-DIM-SM
 ‘short ~ very short’

But in the case of the adjective *gor-ə̌k*, the diminutive cannot attach to the putative root *gor*, as shown in (27).

- (27) **gor-ə̌k** — ***gor-ič-ə̌k**
 poor-SM — poor-DIM-SM

Rather, it must attach after *gorə̌k*, as in (28).

- (28) **gorə̌k** — **gorə̌k-ič-ə̌k**
 poor — poor-DIM-SM

This suggests that *gorə̌k* – even though originally bimorphemic and equipped with an ending that is easily segmentable as a stem marker – is currently treated as an unsegmented root in Bulgarian.

These observations lead us to conclude that the AAB pattern (hard to diagnose for root suppletion) has become the only pattern of stem-marker allomorphy for *-ə̌k* in Bulgarian.

The status of AAB as an active pattern in Bulgarian is supported by the fact that it has been extended beyond the set of adjectives that originally had it. To see this, consider the adjective meaning ‘small’ in (29). In Czech and Ukrainian, this is a suppletive adjective with an ABB pattern that has no stem marker. The Czech pattern is shown in the left part of the table. The right part shows that Bulgarian has extended the stem marker *-ə*^k to accompany this adjective (which does not have it in other Slavic languages). Interestingly, as it adopted the stem marker, it adopted it with the AAB pattern of distribution.

(29) Adjective ‘small’ in Czech and Bulgarian¹⁶

CZECH			GLOSS	BULGARIAN		
POS	CMPR	VERB		POS	CMPR	VERB
<i>mal-ý</i>	<i>men-š-í</i>	<i>z-men-š-i-t</i>	‘small’	<i>mal-ə</i> ^k	<i>po mal-ə</i> ^k	<i>na-mal-i</i>

In sum – and repeating an observation we made above – Bulgarian differs strikingly from its relatives Czech and Ukrainian in the patterning of both the stem marker and the root in the comparative and (causative) change-of-state verbs. While all three languages conform to the CΔG, they do so in different ways. In Czech and Ukrainian, the comparative patterns with the verb (ABB, AAA), in Bulgarian, it patterns with the positive (AAB, AAA). All these patterns are consistent with the CΔG, but we are left with a new question about why the languages should differ so sharply in the patterns they display. Before presenting our analysis, we add data from Turkish which, though unrelated, is strikingly similar to Bulgarian.

3.2 Turkish

The current section provides a brief discussion of Turkish. The description builds on Tabak and Caha (2025) and provides further support for the existence of AAB patterns in stem distribution under exactly the same circumstances as in Bulgarian: with a periphrastic pre-marker in the comparative and a suffixal verb.

Table (30) shows four different classes of the triplets positive – comparative – deadjectival verb. The first line depicts a morphologically simplex adjective *kısa* ‘short.’ We can see that the comparative is formed by an independent word that precedes the adjective, namely *daha*. The inchoative is formed by the suffix *-l*, the causative adds an additional *-t*; *mek/mak* is the infinitival marker.

¹⁶It is worth noting that the cognate of the Czech/Ukrainian comparative root for ‘small,’ namely *men*, has been preserved in Bulgarian as a suppletive root, but now only for the diminutive, compare (ia,b). A diminutive based on the regular root is excluded, see (ic).

- (i) a. *mal-ə*^k b. *mən-ič-ə*^k c. **mal-ič-ə*^k
small-SM small-DIM-SM small.DIM-SM
‘small’ ‘very small’

The issue is thus not that Bulgarian has completely lost the suppletive root. The root was kept, but eliminated in the comparative.

(30) Four classes of deadjectival verbs in Turkish

GLOSS	POSITIVE	COMPARATIVE	INCHOATIVE	CAUSATIVE
short	kısa	daha kısa	kısa -l-(mak)	kısa -l-t-(mak)
small	küçük-k	daha küçük-k	küçük -l-(mek)	küçük -l-t-(mek)
broken	kır-ık	daha kır-ık	kır -ıl-(mak)	kır -(mak)
far	uza-k	daha uza-k	uza -(mak)	uza -t-(mak)

The last three rows depict adjectives that are morphologically complex in the positive degree, and feature a stem marker **-(I)k**. We know that this is an independent morpheme because (just like in Slavic) the diminutive comes in between the root and the stem marker, as in (31) (see Türk 2020, Atlamaz and Dikmen 2026 for discussion).¹⁷

- (31) **küçük-k** ~ **küçük-cü-k**
 small-SM small-DIM-SM
 ‘small’ ~ ‘very small’

The crucial property of the stem marker is that it is found in the positive and in the (prefixal, analytic) comparative, but never in deadjectival verbs, just like the Bulgarian **-ə̣k**. This is the case for all the adjectives in the last three rows in Table (30).

The different adjective classes in the last three rows differ in the amount of verbal morphology. The adjective *küçük* ‘small’ shows the ‘regular’ pattern with the inchoative **-l** and the causative **-t**. The adjective *uza* ‘far’ lacks the inchoative **-l**, and finally, the root *kır* ‘broken’ lacks any verbal morphology in the causative.

Regardless of how these three different classes are analyzed (see Tabak and Caha 2025 for a proposal), the fact remains that all of them show an AAB pattern of stem distribution.

The following table provides a number of different roots exhibiting the AAB pattern, with the stem marker **-(I)k** in the positive and in the comparative, but not in the deadjectival verb. The roots are divided into two classes, depending on whether they have **-l** in the inchoative or not.¹⁸

¹⁷Atlamaz and Dikmen (2026) analyze the diminutive **-cük** as a single morpheme. This does not change our main point, which is that in the positive adjective, the final **-(I)k** is a stem marker. In Atlamaz and Dikmen (2026), **-cük** simply replaces the adjectival stem marker.

¹⁸Those that have the **-l** in the inchoative fall into two different classes based on the causative, recall (30).

(32) Adjectives exhibiting the AAB pattern (Tabak and Caha 2025)

	POS	CMPR	INCHOATIVE
low	alça-k	daha alça-k	alça -l
tiny	mini-k	daha mini-k	mini -l
little	ufa-k	daha ufa-k	ufa -l
small	küçü-k	daha küçü-k	küçü -l
broken/rotten	boz-uk	daha boz-uk	boz -ul
soft	yumuşa-k	daha yumuşa-k	yumuşa
far	uza-k	daha uza-k	uza
big	büyük-k	daha büyük-k	büyük
cold	soğ-u-k	daha soğ-u-k	soğ-u
flexible	esne-k	daha esne-k	esne

This concludes our look at Turkish. The data strengthen the conclusion reached for Bulgarian that there is a correlation between the AAB pattern and a prefixal/periphrastic marking of the comparative plus suffixal marking of the change-of-state verb.

3.3 Bulgarian redux

Before proceeding to the Nanosyntax and DM analyses of these patterns, we make a brief digression to call attention to one way in which Bulgarian and Turkish are even more parallel than they may seem. In both languages, the comparative marker is a pre-element: in Turkish a separate function word *daha*, in Bulgarian the element *po* described as a prefix. However, as has been noted elsewhere, *po* seems to exhibit some properties of an independent word. One of these properties is shown in (33). This example demonstrates that it is possible to coordinate two adjectives in the scope of a single comparative marker, very much like in the English translation *more [interesting and entertaining]*.¹⁹

- (33) Tazi kniga beše **po-** [interesn-a i zabavn-a] od predišna-ta
 this piece was more interesting and entertaining from previous.DEF
 ‘This book was more interesting and entertaining than the previous one.’

The Czech example in (34) shows that the Czech comparative marker must be repeated on each conjunct. It would be impossible to delete one of them, while maintaining the comparative reading of both adjectives.

¹⁹We remain neutral with respect to the analysis, i.e., it can either be the case that two APs are coordinated in syntax and embedded under a single CMPR, or it can be that there is some kind of a deletion involved, as in (i).

- (i) a. CMPR [ADJ & ADJ]
 b. [CMPR ADJ] & [~~CMPR~~ ADJ]

What is important for us is that there is a contrast with Czech, which we turn to immediately.

- (34) Tato kniha byla [zajímav-ějš-í] a [zábavn-ějš-í] než ta předchozí
 This book was interesting-CMPR and entertaining-CMPR than the previous
 ‘This book was more interesting and entertaining than the previous one.’

Another property that sets the Bulgarian *po* apart from the Czech suffixal marker is that Bulgarian *po* has less strict selectional restrictions. In (35), we illustrate this with locative adverbs. In (35a), we show the locative adverb *gore* meaning ‘up, on top.’ In (35b), the locative is preceded by a goal-path marker *na*, with the combination *na gore* meaning literally ‘to (a place which is) up.’ The resulting complex goal expression *na gore* can be preceded by *po* in the meaning ‘more.’

- (35) Bulgarian
 a. Toj e **gore** na stǎlba-ta
 he is up.LOC on ladder-the
 ‘He is up on the ladder.’
 b. Kateri se (**po**) na **gore**!
 climb REFL CMPR to up
 ‘Climb (higher) up!’

In (36), we turn to Czech. In (36a), we show that to get the ‘more’ reading of the cognate directional in Czech, we can use the analytic marker *víc* ‘more.’ In (36b), we show that one cannot use the suffixal comparative *-ěji* to express such meaning.

- (36) Czech
 a. Vylez (**víc**) na-horu
 climb more on-up
 ‘Climb (higher) up!’
 b. *Vylez na-hoř-**ěji**
 climb on-top-CMPR

This supports the idea that the Bulgarian comparative *po-* is an analytic marker with looser selectional restrictions than the suffixal comparative in Czech.

Finally, there is also a phonological property that sets the comparative *po* apart from prefixes. (37a) shows that the comparative *po-* is always stressed, and the material to which it attaches keeps its own word stress (Leafgren 2011, 42). In this example, stress is marked by an acute accent over a vowel (ó). (37b) shows that the stress pattern is different for the homophonous verbal prefix, which does not bear stress.²⁰

- (37) a. nów ~ pó- nów
 new CMPR new
 ‘new ~ newer’
 b. dám ~ po-dám
 give ON-give
 ‘give ~ pass’

²⁰The suppletive quantifier ‘more’ *póveče* only has a single stress (compare *pó-nów* ‘newer’), suggesting potentially a monomorphemic, non-decomposable nature of this item. The orthography reflects this, *poveče* is written as one word, *po-nov* with a hyphen. We come back to this in Section 6.

3.4 Interim summary: AAA, ABB, AAB

In our limited sample, Bulgarian and Turkish differ from Czech and Ukrainian along four dimensions. In the first place: (i) Czech and Ukrainian show lexical idiosyncrasy in whether the stem-marker is retained or lost in the comparative; and (ii) this irregularity in the comparative is (with some caveats) inherited by the corresponding change-of-state verbs, yielding AAA and ABB patterns, but not AAB patterns. Bulgarian and Turkish show (i) no irregularity in the comparative – the stem marker is always retained, and (ii) the change-of-state verbs systematically fail to inherit the comparative form, giving AAB patterns, but not AAA or ABB (with one caveat). Two additional ways in which these languages differ are that (iii) the comparative marker follows the adjectival stem in Czech and Ukrainian, but precedes it in Bulgarian and Turkish, and (iv) the construction is synthetic (affixal) in Czech and Ukrainian but analytic in Turkish and arguably Bulgarian.

These observations amount to the AAB conjecture, originally (9), repeated in (38) for convenience.

- (38) The AAB conjecture
Systematic AAB patterns arise in languages with analytic, pre-marking comparatives and suffixal change-of-state verbs.

Our goal in the following sections is to provide Nanosyntactic and DM accounts that will relate these differences, aiming to derive distinctions (i) and (ii) from (iii) or (iv). To anticipate the outcome, we note that the groundwork for this has already been set regarding the domain of suppletion, such that the patterns described above mostly follow from our starting assumption that allomorphy in stem markers should be treated the same as root allomorphy.

The starting point is the observation (Bobaljik 2012, 70–71) that quite generally, when languages have an alternation between a synthetic (suffixal) construction and a periphrastic one (typically with pre-marking), suppletion is obligatory in and/or limited to the synthetic one. Thus English *go* has a suppletive form *went* in the past, but only when these are synthetic – the periphrastic past (needed in *do*-support contexts such as negation and non-subject questions) is non-suppletive: *did go* not **did wen(d/t)*.

The table in (39) presents a few examples of this alternation from comparative formation. In each of these languages, analytic pre-markers always combine with the form of the positive (exactly as in Bulgarian), while analytic forms with a (sometimes null) suffix undergo suppletion:²¹

²¹See Bobaljik (2012, 70–71) for qualifications; note that some languages (at least colloquially) allow for the pre-marker to combine with a morphological comparative (with or without suppletion) apparently as a reinforcement of the comparative sense. In some (French, Italian) it is prescriptively considered an error (pleonasm), but in others (Greek) it is acceptable.

- (39) Suppletion is limited to suffixal forms (Bobaljik 2012, 70–71)

LANGUAGE	POS	SUFFIXAL CMPR	SYNTHETIC PRE CMPR	GLOSS
French	<i>mauvais</i>	<i>pire</i>	plus <i>mauvais</i>	‘bad’
Italian	<i>buon-o</i>	<i>miglior-e</i>	più <i>buon-o</i>	‘good’
Greek	<i>kak-ós</i>	<i>cheiró-ter-os</i>	pjo <i>kak-ós</i>	‘bad’

On the basis of these facts, Bobaljik (2012, 3) has proposed the generalization in (40):

- (40) The Root Suppletion Generalization (RSG)
 Root suppletion is limited to synthetic (i.e., morphological) comparatives.

However, the facts in (39) are, in and of themselves, compatible with an alternative interpretation, which is that it is the pre-marker status of the morphemes that prevents suppletion from taking place. This line of reasoning was followed by De Clercq and Vanden Wyngaerd (2018, 342), who propose the generalization in (41):

- (41) Generalization on Suppletion and PRE marking (GOSP)
 When there is root suppletion, the marker of the comparative degree cannot occur [exclusively] to the left of the adjectival root.²²

Both of these generalizations extend beyond root suppletion to cover the differences in stem-marker distribution that are the main focus of this paper. We now turn to showing how Nanosyntax and DM derive (41) and (40), respectively.

4 Nanosyntax account: pre-markers as ‘last resort’

This section presents an account of how the facts observed up to now can be modeled within the Nanosyntax approach (Starke 2009, De Clercq et al. 2025). Previous explorations of degree morphology within Nanosyntax argued for a rich underlying structure.²³ To allow for a cross-framework comparison, much of the complexity unearthed in the previous studies will be ignored, and the account uses the structures as presented in (5)–(7). The main analytical goal will be to show how ABB arises in a suffixal language, while in a prefixal language, AAB is derived.

The core of the explanation draws on an established property of the Nanosyntax framework, which is that in syntactic derivations containing a prefix, the prefix is formed in a separate workspace, whereas suffixation involves only a single workspace. Even though the opening of a separate workspace reflects only the prefix status, it ultimately links the approach to a notion of locality, which forms the core of the DM account formulated in Section 5: markers in two separate workspaces seem to

²²We added the word *exclusively* into the generalisation to cover cases where a pre-marker reinforces a suffixal comparative (e.g., in the Greek *pjó cheiro-ter-os*, literally *more worse*).

²³Specifically, it has been argued that comparative markers in Czech and Ukrainian show various allomorphs that require multiple comparative heads (Caha et al. 2019, Vyshnevskaya 2022, Caha et al. 2024). It has been also observed that both Czech and Ukrainian have several different stem markers. This led to analyses with multiple heads within the positive degree (Vanden Wyngaerd et al. 2020, Vanden Wyngaerd et al. 2025, Vyshnevskaya 2025b; see also Wiland 2023 for Polish, Kasenov 2025 for Russian).

have more restricted options of influencing each other than markers within a single workspace.

The section is structured as follows. Section 4.1 introduces the core assumptions needed to derive the facts. Section 4.2, sketches an approach to ABB patterns in suffixing languages, with an extension to root suppletion patterns in Section 4.3. Section 4.4 provides an account of AAB patterns in prefixing languages like Turkish or Bulgarian. Section 4.5 concludes.

4.1 The assumptions

This section assumes that the reader is familiar with the basic tenets of Nanosyntax as outlined in introductions such as De Clercq et al. (2025), Caha (2026a) or Caha (2026b). Against this background, it lays out the specific ingredients of the framework that are needed to derive the AAB Conjecture.

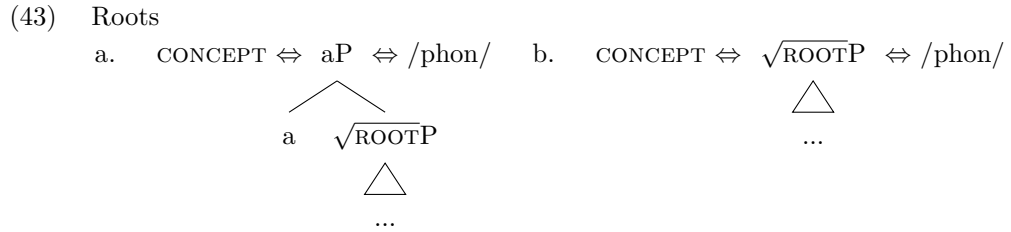
The first assumption concerns the distinction between morphologically simplex and complex adjectives. The analysis of these two classes assumed here is shown in Table (42).

(42)

	$\sqrt{\text{ROOT}}$	a
morphologically simplex adjective	root	
adjective with a stem marker	root	-(V) k

The header of the table lists the relevant features that need lexicalization in the positive ($\sqrt{\text{ROOT}}$ and a). The idea depicted on the first row is that morphologically simplex adjectives lexicalize both structural ingredients. The last row depicts complex adjectives, where the root morpheme lexicalizes only the $\sqrt{\text{ROOT}}$ part of the structure and needs the stem marker to lexicalize little a . The stem marker is represented as $(V)k$ in the table, as it conveniently corresponds to a vowel followed by k in all the languages.

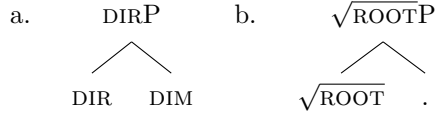
The type of lexicalization chosen has to do with the lexical entry of each root. When the root is lexically linked to a ‘large’ constituent containing the $\sqrt{\text{ROOT}}$ and the little a , as in (43a), the root lexicalizes the full structure of the positive.



With complex adjectives, the root lexicalizes just the $\sqrt{\text{ROOT}}$ part (and the stem marker is needed to lexicalize little a). To express this formally, some footwork is needed. The crucial point is that in Nanosyntax, lexicalization only targets phrasal nodes. Therefore, the lexical item in (43b) assumes that the $\sqrt{\text{ROOT}}$ node in the positive (5) has complex internal structure. There are two ways in which this idea can

be implemented. One is that the root constituent contains two regular features. The current Nanosyntax work on the structure of adjectives (Vanden Wyngaerd et al. 2020, Wiland 2023 Vyshnevskaya 2025a) calls these DIMENSION (e.g., SPEED) and an ordering along this dimension, referred to as DIRECTION. The direction can be positive (yielding an adjective such as *fast*) or negative (yielding the antonym related to the same dimension, i.e., *slow*). Under this approach, the $\sqrt{\text{ROOTP}}$ in (43b) would correspond to the structure in (44a).

(44) Two ways to think about the complex $\sqrt{\text{ROOTP}}$



In order to provide as minimal comparison to DM as possible, we follow here a different track, depicted in (44b). On this approach, the first merge always involves the merger of a feature with an empty set, represented in (44b) by a dot (see De Belder and Van Craenenbroeck 2015).

We follow the standard Nanosyntax assumption that syntactic derivation proceeds in cycles, where cycles are defined by external Merge. When a new feature *F* is merged, the resulting FP must be ‘externalizable,’ i.e., there must be a lexical item that matches it.²⁴ If there is a matching lexical item for the FP created, the structure is legible at the interface. In such case, the derivation may continue by adding a new feature (or terminate if all required features have been merged). When there is no match, syntax performs rescue operations as described in (45b-e).²⁵

(45) The lexicalization algorithm

- a. Merge *F* and lexicalize.
- b. If a. fails, evacuate the closest labeled constituent that (a) contains the root and (b) has no single-branching nodes inside.
- c. If b. fails, pied-pipe the immediately dominating node and lexicalize.
- d. If c. fails, go back to the previous cycle, and try the next option for that cycle.
- e. If d. fails, spawn a new derivation providing feature *X* and merge that with the current derivation, projecting feature *X* to the top node.

First, syntax tries to evacuate out of FP the closest constituent with the relevant properties, see (45b). If this does not work, it pied-pipes larger and larger constituents, see (45c).

Since all movements are to the left, derivations with evacuation movements always move a constituent containing the root to the left of FP. The result is that evacuation

²⁴Matching between entries and structures is governed by the condition in (i).

(i) *Condition on matching* (De Clercq et al. 2025, 12): A lexically stored constituent *L* matches a syntactic phrase *S* iff *S* is identical to *L*.

²⁵The architecture of the algorithm comes from Starke (2018), but includes later modifications, for which see De Clercq et al. (2025) and Wiland (2026).

movements lead to suffixes (the root is to the left of whatever lexicalizes the FP). Prefixal derivations are only tried when all evacuations fail. So, in the very last option of the algorithm, a new derivational workspace opens where a new phrase marker is constructed, see (45e). In the course of the discussion, we will show that if prefixation is the very last step of the algorithm, this leads to the result that comparative prefixes will always attach to the form of the positive. The backtracking step (45d) will be discussed as the discussion unfolds.

4.2 ABBs in suffixing languages

This section shows one possible way of deriving ABB patterns in Nanosyntax. Specifically, we provide a step-by-step derivation of the two examples depicted in (46).

(46)

language	gloss	positive	comparative	inchoative
Ukrainian	sweet	solod- k -yj	solod- š -yj	solod- š - a -ty
Czech	tame	krot- k -ý	krot- š -í	krot- nou -t

A preview of the analysis for Czech is provided in the upper part of the lexicalization table (47). The features of the individual forms are on the top row. Black color in the table means that the feature is not present in the relevant form. We can see that the root always lexicalizes the $\sqrt{\text{ROOT}}$ part, and the suffix (the stem marker *-k*, the comparative *-š* and the inchoative *-nu*) lexicalize the remaining projections.

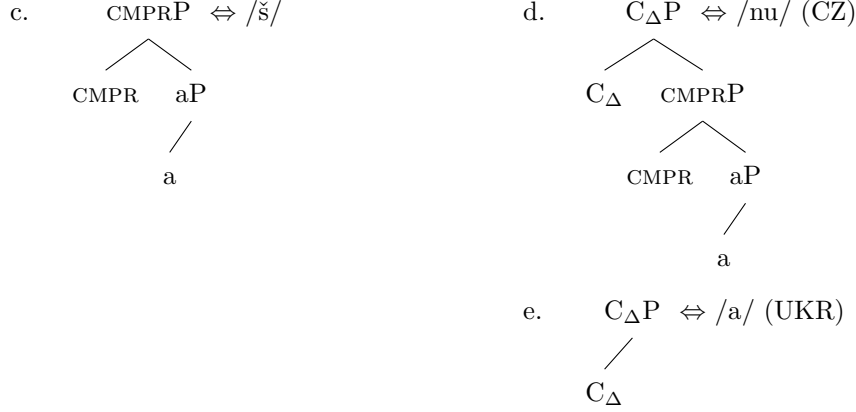
(47)

		$\sqrt{\text{ROOT}}$	a	CMPR	C_Δ
CZ	positive	krot	-k		
	comparative	krot	-š		
	inchoative	krot	-nu		
UKR	positive	solod	-k		
	comparative	solod	-š		
	inchoative	solod	-š		-a

The analysis for the Ukrainian pattern is shown in the bottom part of the table. The crucial difference is that Ukrainian does not have the inchoative suffix *-nu*, and splits the features of the inchoative across two suffixes, namely *-š* and *-a*. These lexicalizations arise in Nanosyntax through the interaction of the Lexicalization algorithm (45) and the lexical entries given in (48).

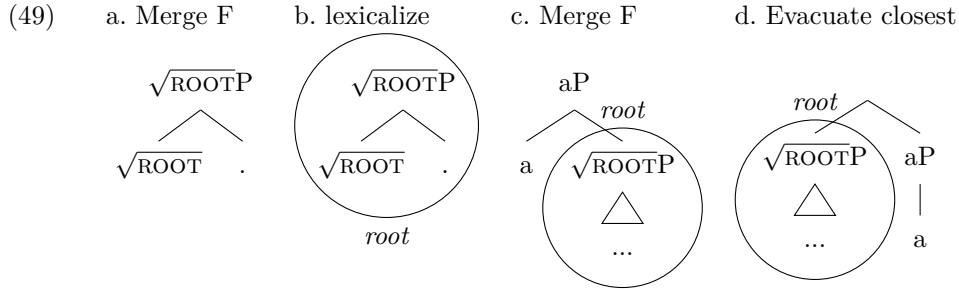
(48)

a.	$\sqrt{\text{ROOTP}} \Leftrightarrow /krot \sim solod/$	b.	$aP \Leftrightarrow /k/$
	$\triangle$		$/$
	...		a



The entry for the root, for the stem marker, and for the comparative marker are the same for both languages, see (48a-c). The inchoative *-nu* is only found in Czech, see (48d), while the inchoative *-a* is only found in Ukrainian, see (48e).

We now proceed to show how the entries in (48) yield the lexicalization as depicted in Table (47). The derivation starts the same in both languages. The $\sqrt{\text{ROOTP}}$ is constructed, as in (49a). The root entries in (48a) are a perfect match, and therefore, lexicalization takes place, see (49b). Since matching is successful, the syntactic object is legible at the interface and the derivation can continue by Merge F. When little *a* is added, as in (49c), there is no match. Since the structure is not legible at the interface (there is no way to map it onto PF), the phrase marker is rejected at the interface and handed over to syntax for rescue movements.

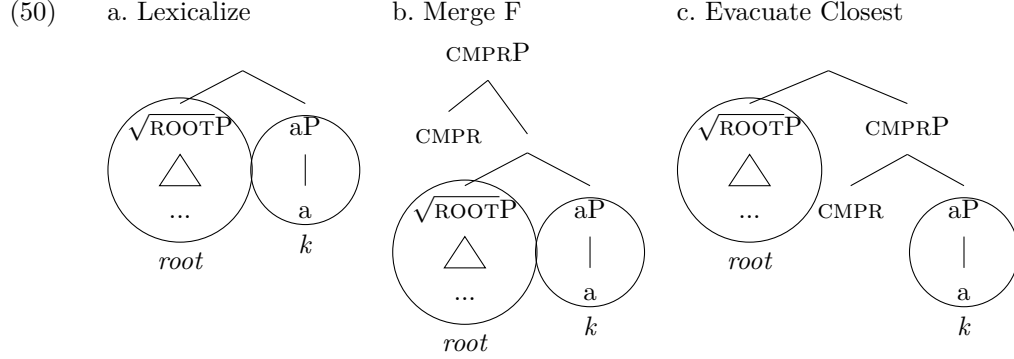


Syntax now evacuates from within *aP* the closest labeled constituent that has the right properties, namely, it contains the root and has no single branching nodes inside it. That happens to be the $\sqrt{\text{ROOTP}}$ immediately below *aP*, with the result of the evacuation movement given in (49d).²⁶

In this structure, the $\sqrt{\text{ROOTP}}$ is still matched by the root, and the difference to (49c) is that in (49d), *aP* has only one daughter. As a result, the lexical entry of the

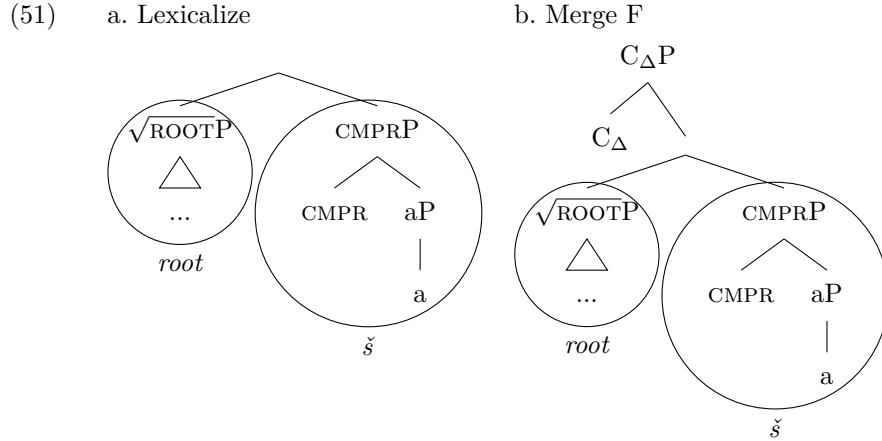
²⁶Note that Nanosyntax adopts the convention that the trace of this kind of ‘lexicalization’ movement is ignored, and the top node has no label (see De Clercq et al. 2025).

stem marker $-k$ as given in (48b) is identical to this constituent and lexicalization takes place, see (50a).



When CMPR is added, as in (50b), there is no match for the FP created, i.e., no match for CMPRP. Therefore, syntax has to evacuate the closest labeled constituent that has the root and no single-branching nodes inside, which is again $\sqrt{\text{ROOT}}\text{P}$. The structure that arises after $\sqrt{\text{ROOT}}\text{P}$ evacuates is shown in (50c) (the ‘traces’ of $\sqrt{\text{ROOT}}\text{P}$ are ignored, and the position of the root in between CMPRP and aP is thus eliminated).

In the structure (50c), the comparative marker $-\check{s}$ as given in (48c) perfectly matches the (remnant) CMPRP. Therefore, we get a successful lexicalization depicted in (51a). This is the correct comparative, corresponding to the root followed by $-\check{s}$ in both languages. Importantly, the stem marker $-k$ no longer has a role to play in the comparative, so, when it comes to the distribution of this marker, we have correctly derived the start of an **ABB** pattern.²⁷



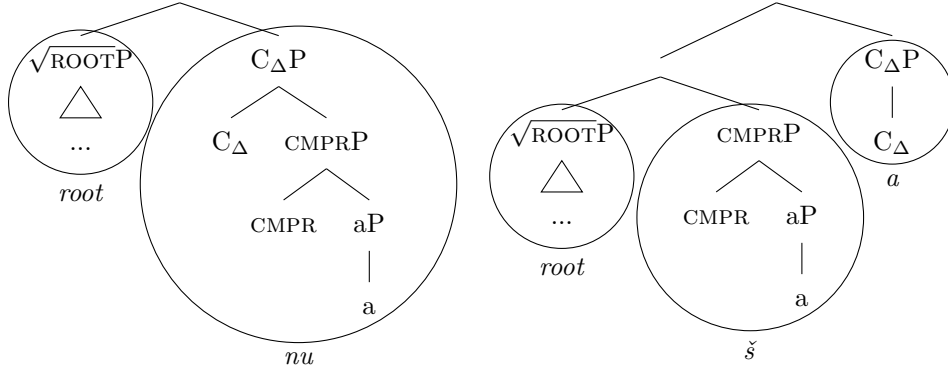
The final step is to merge the C_Δ feature, producing (51b). Here the top node has no match, therefore, evacuate closest produces the structure in (52a). In Czech, this

²⁷The derivations sketched here are simplified compared to existing Nanosyntax accounts, which also address the issue of how the stem marker is retained with other roots; see Vanden Wyngaerd et al. (2025) or Vyshnevskaya (2025a).

structure is realized as the root followed by *-nu*, since the lexical entry for *-nu* in (48d) provides a perfect match for the remnant $C_{\Delta}P$ in (52a). This is the correct structure for Czech: with no *-k* and hence, the full ABB pattern successfully emerges.

(52) a. Evacuate Closest

b. Pied-pipe



In Ukrainian, there is no match for $C_{\Delta}P$ in (52a). Therefore, the ‘Evacuate Closest’ step is undone (it led to no result), and the derivation returns to the point where C_{Δ} was merged, i.e., (51b). The next evacuation step pied-pipes the node immediately above $\sqrt{\text{ROOTP}}$, i.e., the whole complement of C_{Δ} , producing (52b). Here the remnant $C_{\Delta}P$ is matched by *-a*, producing the correct form for Ukrainian, namely *solod-š-a*.

Summarizing, this section showed how an ABB pattern of stem allomorphy arises in Nanosyntax. The stem marker is inserted to lexicalize a particular projection, little *a*, but disappears in the other forms because the comparative *-š* or the inchoative *-nu* make it unnecessary: they themselves lexicalize the projection for which the stem marker was needed in the first place.

4.3 Root patterns: AAA and ABB

This section shows that a virtually identical derivation as that presented in Section 4.2 is also responsible for ABB and AAA patterns in morphologically simplex adjectives. To show that, we focus on the pattern found with the Ukrainian adjectives ‘bad’ and ‘young’ as given in Table (53). The adjective ‘bad’ is suppletive, showing an ABB pattern. The adjective *molod* ‘young’ has an AAA pattern.

(53)

gloss	adjective	comparative	inchoative
bad	pohan -yj	hir -š-yj	hir -š-a-ty
young	molod -yj	molod -š-yj	molod -š-a-ty

The gist of a Nanosyntax analysis is presented in Table (54).

(54)

ROW		$\sqrt{\text{ROOT}}$	a	CMPR	C_{Δ}
1	positive ‘sweet’	solod	-k		
2	comparative	solod	-š		
3	inchoative	solod	-š		-a
4	positive ‘bad’	pohan			
5	comparative	hir	-š		
6	inchoative	hir	-š		-a
7	positive ‘young’	molod			
8	comparative	molod	-š		
9	inchoative	molod	-š		-a

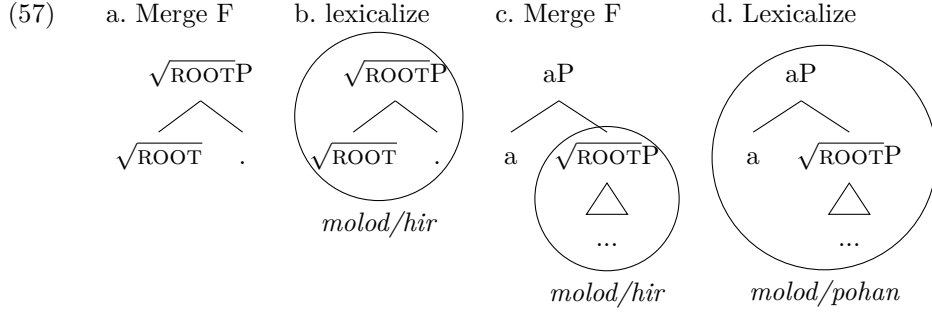
The table shows that the main difference between the complex adjective ‘sweet’ and the simplex adjectives is in the positive degree. While ‘sweet’ requires two morphemes to lexicalize the positive (see Row 1), adjectives like ‘bad’ or ‘young’ only show the root morpheme in the positive, see Rows 4 and 7. The remaining forms of ‘bad’ and ‘young’ copy the same lexicalization structure as the adjective ‘sweet,’ i.e., the comparative has an analogous lexicalization across all the forms, compare Rows 2, 5 and 8.

The analysis depicted in (54) proposes that the root of morphologically simplex adjectives like ‘bad’ or ‘young’ lexicalizes different projections in the positive vs. in the comparative and the change of the state verb. In the table, the ABB pattern of root suppletion arises when the differing sets of heads have each a dedicated lexical item: *pohan* for $[\sqrt{\text{ROOT}}+a]$, *hir* for just the $\sqrt{\text{ROOT}}$; see Rows 4-6. The entries for the two roots are given in (55), both linked to the same concept.²⁸

- (55) a. $\text{BAD} \Leftrightarrow \text{aP} \Leftrightarrow /pohan/$
- $\begin{array}{c} \diagup \quad \diagdown \\ \text{a} \quad \sqrt{\text{ROOTP}} \\ \triangle \\ \dots \end{array}$
- b. $\text{BAD} \Leftrightarrow \sqrt{\text{ROOTP}} \Leftrightarrow /hir/$
- $\begin{array}{c} \triangle \\ \dots \end{array}$
- (56) $\text{YOUNG} \Leftrightarrow \text{aP} \Leftrightarrow /molod/$
- $\begin{array}{c} \diagup \quad \diagdown \\ \text{a} \quad \sqrt{\text{ROOTP}} \\ \triangle \\ \dots \end{array}$

The adjective ‘young,’ shown on the last three rows of Table (54), exhibits the same lexicalization structure as ‘bad.’ The difference is that its root is syncretic/ambiguous between two different sizes: aP and $\sqrt{\text{ROOTP}}$. To see how this works, consider the entry for *molod* ‘young’ in (56). The ambiguity arises because the root has a constituent that is identical to $\sqrt{\text{ROOTP}}$ and also to aP. Based on the matching condition, this entry allows the root to lexicalize both constituents. Let us now go through the derivation in detail. The derivation starts by assembling $\sqrt{\text{ROOTP}}$, see (57a).

²⁸Nanosyntax standardly uses pointers to model root suppletion, see Vanden Wyngaerd et al. (2021). We do not introduce pointers here for simplicity.

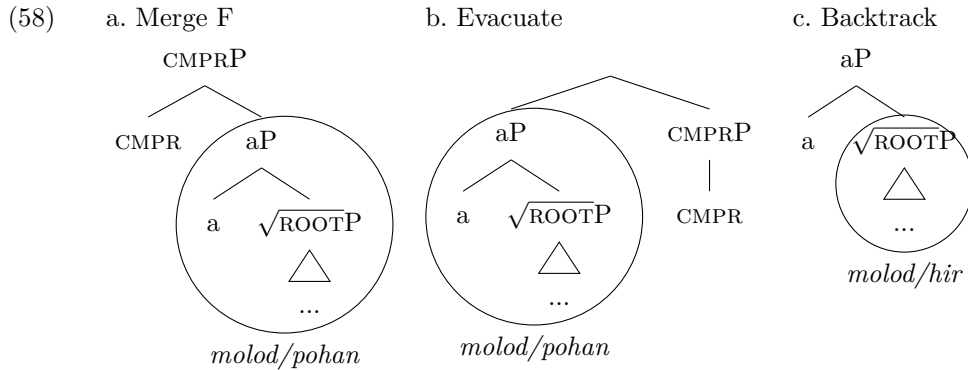


All three roots in (55) and (56) have a constituent that is identical to the structure in (57a), therefore, all are candidates for insertion. Now when it comes to roots that are associated to different concepts, Nanosyntax assumes that we chose the root that is associated to the right concept, i.e., the one that we ‘want to talk about.’ So, the adjective *molod* ‘young’ can be inserted, as in (57b), as long as that is what we want to talk about.

However, this logic does not apply in the case of the roots *pohan/hir*, as they are both linked to the same concept. In such case, the root *hir* is chosen based on the Elsewhere Condition, as it has fewer superfluous features.

When little *a* is merged, deriving the positive, we get the structure in (57c). Here *molod* ‘young’ is still a match, but *hir* ‘bad’ is not. Hence, for this concept, the grammar can only use *pohan* ‘bad,’ which is the correct result for the positive degree.

If the positive was to be derived, the derivation of the adjective would terminate. If we want to form a comparative, the comparative feature is added, as in (58a).



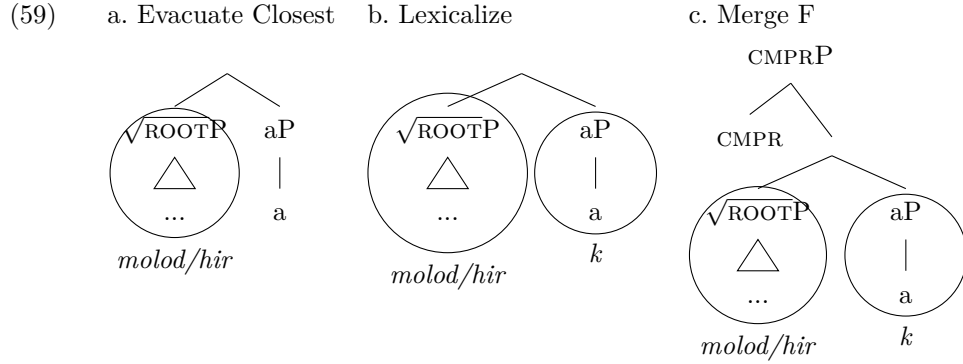
There is no entry that matches the CMPRP, therefore, the closest labeled constituent containing the root must be evacuated out of it. This yields (58b). However, there is no match for CMPRP in (58b) either, as none of the entries in the Ukrainian (or Czech) lexicon contains a CMPRP with a single daughter.

Therefore, moving to the next step of the algorithm, the constituent immediately dominating aP must be now extracted out of CMPRP. However, the node immediately dominating aP is the CMPRP itself, so pied-piping does not lead to any new structure. Therefore, we move to the next step of the algorithm, which is Backtracking.

Backtracking has the following logic: since adding CMPR to the positive degree as in (58a) failed to produce any output, let us go back to the positive, try to do something differently and then merge CMPR to a different structure, hoping that now, lexicalization will succeed. Backtracking thus goes back to the stage when we added little *a*, see (58c). Notice that as we go back, we fall back on the root *hir* ‘bad’ as the lexicalization of $\sqrt{\text{ROOTP}}$.

On the first attempt, we lexicalized aP without any movements, so now we try to evacuate closest, see (59a). Here, the remnant aP can be lexicalized by the stem marker, see (59b). We thus derive a different form of the positive, one which includes the stem marker. However, this is not a wrong result, as this form only exists as an intermediate step of deriving the comparative, which is derived by adding the CMPR head on top of (59b), yielding (59c).

Here the derivation reaches the same form as in the derivation of complex adjectives, recall the (b) step in (50). Therefore, the derivation now follows exactly the same steps, deriving the comparatives *molod-š-* ‘younger’ and *hir-š-* ‘worse.’ When the C_{Δ} head is added, the correct inchoative verb (*hir-š-a-ty* and *molod-š-a-ty*) also fall out in the same way as in Section 4.2.



Summarizing, this section showed how the ABB pattern arises for suppletive adjectives. We also showed that the derivations needed to achieve this are compatible with an AAA pattern of marking. These results serve as a backdrop of a discussion of prefixal marking in the next section.

4.4 AAB in prefixing languages

This section turns to languages with prefixal comparatives. The specific case to be discussed is the AAB pattern found with the adjective ‘small’ in both Turkish and Bulgarian, see (60).²⁹

²⁹The verb *na-mal-i* ‘decrease’, which we list in (21) as an inchoative, was listed in (60) as causative. The verb can be used both as a causative and as an inchoative.

(60)

	POS	CMPR	INCHOATIVE
small TURK	küçü-k	daha küçü-k	küçü -l-
small BG	mal-ək	po mal-ək	na-mal -i-

To account for the fact that the relevant adjectives exhibit a stem marker in the positive, we again propose that the root only lexicalizes the $\sqrt{\text{ROOT}}$ part, and the stem marker lexicalizes little *a*, see row 1 in (61). The analysis of the positive is thus the same as in Czech, given on Row 4, repeated from (47).

(61)

ROW	language	gloss	$\sqrt{\text{ROOT}}$	a	CMPR	C_Δ
1	Turkish	small, POS	küçü	-k		
2		small, CMPR	küçü	-k	daha	
3		small, INCH	küçü		-l	
4	Czech	tame, POS	krot	-k		
5		tame, CMPR	krot	-š		
6		tame, INCH	krot		-nu	

The Turkish (and Bulgarian) comparatives differ from Czech in that the stem markers are preserved. We therefore propose that the Turkish comparative *daha* (and similarly the Bulgarian *po*) only lexicalize CMPR. As a result, the whole positive is preserved in the comparative, including the stem marker, see Row 2 (and compare the Czech Row 5, where the stem marker is eliminated).

Note as well that since the comparative marker *daha* comes on top of the full positive in the lexicalization table, there can be no root suppletion with simplex adjectives: the root must lexicalize precisely the same heads in the positive and in the comparative, namely $\sqrt{\text{ROOT}}+a$.

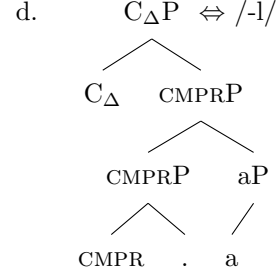
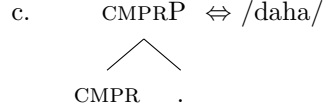
Finally, we propose that the Turkish inchoative *-l* (and the Bulgarian labile *-i*) lexicalize the very same projections as the Czech *-nu*, leading to the disappearance of the stem marker *-k*, and hence, to an AAB pattern.

The important point that emerges from (61) is that Czech and Turkish run in parallel except in the comparative, where the prefixal marker *daha* lexicalizes just CMPR. This in turn leads to the preservation of the stem marker and to the absence of suppletion. Thus, the prefix-suffix asymmetry suggested by the data can now be restated as follows: prefixes (like the CMPR prefix) never lexicalize features below the level at which they are introduced. The reasons why such derivations are impossible will be addressed after we highlight the actual derivation.

We start by presenting the lexical items needed for the Turkish pattern in (62).

(62)

a.	$\sqrt{\text{ROOT}}P \Leftrightarrow /küçü/$	b.	$aP \Leftrightarrow /k/$
	$\triangle$		$/$
	...		a

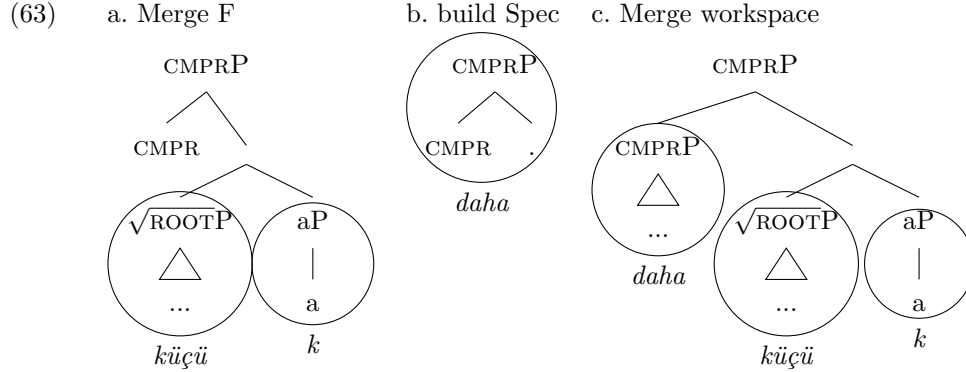


The entry for the root and for the stem marker are the same in Turkish and Bulgarian as in Czech and Ukrainian. This allows us to assume a diachronic continuity of these markers in Bulgarian, locating the difference inside the prefix instead.

The difference between the two sets of languages is that an entry for the suffixal comparative is missing. Instead, we have the prefixal markers *daha/po*. The difference between prefixal and suffixal markers is that suffixal markers always have a unary-branching phrasal node at the bottom with just a single daughter. Such entries can only be used when their complement moves to the left. The comparative *daha* is different: its CMPRP has two daughters, and so there can be no evacuation of the complement. In Nanosyntax, phrases like these are constructed in a separate workspace by merging CMPR with the empty set, and only after the phrase is formed, it is attached to the main projection as a complex left branch (Starke 2018).

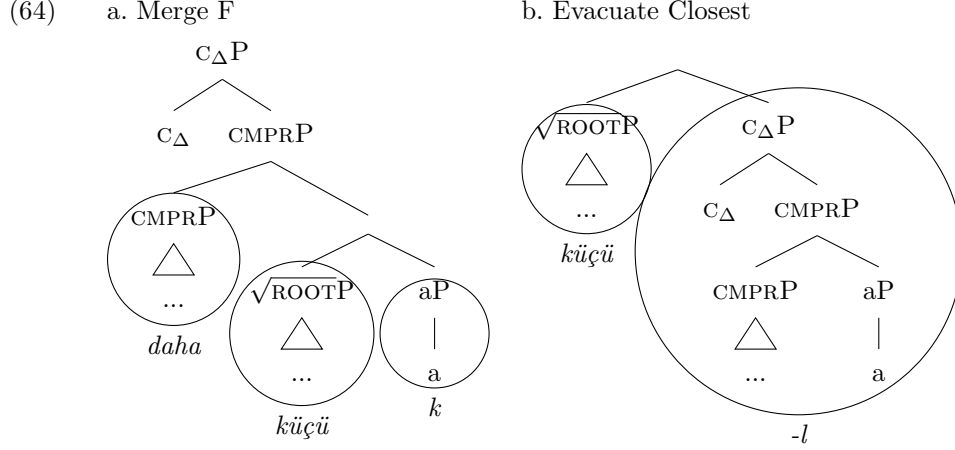
Finally, the inchoative *-l* is slightly different from the Czech *-nu* (recall (48d)), in that it has a phrasal CMPRP in between C_{Δ} and little *a*, but otherwise it lexicalizes the same features and also has a unary bottom. It is therefore a suffix, which – due to its ability to lexicalize little *a* – makes the stem marker *-k* unnecessary.

The derivation is sketched below. The initial steps – leading to the positive with a stem marker – are not shown, as they are the same as in Czech or Ukrainian complex adjectives, recall Section 4.2. We start following the derivation when the CMPR head is introduced, see (63a). At this point, no matter what evacuation movement out of CMPRP we try, there will be no match for it, simply because there is no suffixal entry containing CMPR as a simple feature.



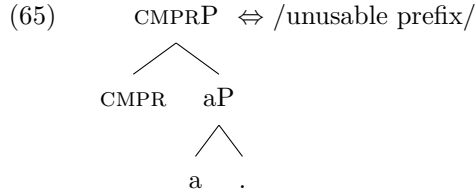
Therefore, after all evacuations have been tried, including backtracking, suffixal lexicalization fails. The derivation then returns to the stage where CMPR was merged as a feature, see (63a). The feature is removed and a new workspace is opened. Here CMPR is merged with an empty set (as is the case with all initial mergers). The resulting phrase is in (63b), and it can be lexicalized as *daha*. The phrase is then merged with the main workspace, yielding the correct comparative, see (64c).

When C_{Δ} is introduced, as in (64a), it is impossible to lexicalize $C_{\Delta}P$.



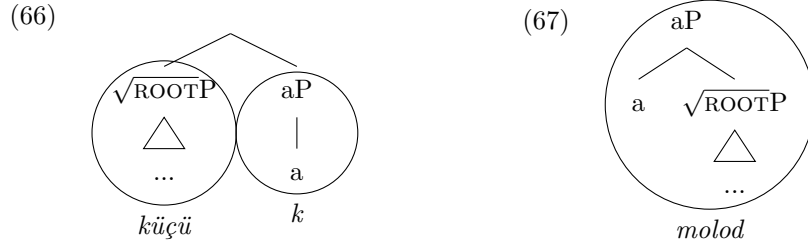
Therefore, evacuations must be tried. Assuming with Wiland (2026) that the closest labeled constituent may not contain single-branching nodes, and must contain the $\sqrt{\text{ROOT}}$, the ‘Evacuate Closest’ operation targets the $\sqrt{\text{ROOT}P}$ in (64a), and produces (64b) (where the trace of the evacuated phrase is eliminated along with the dominating node).³⁰ After evacuation, $C_{\Delta}P$ can be lexicalized by the inchoative suffix, eliminating the stem marker in the process. At this point, we have derived the lexicalization pattern given in (61).

The final question we must address is why the type of derivation attested for suffixes (exemplified by the Czech adjective ‘tame’ in (61)) is not available for prefixes, i.e., why it is the case that there can never be a prefix like the one in (65), that would lexicalize little *a* along with CMPR, thereby causing the stem marker to disappear.



³⁰On some approaches to ‘Evacuate closest,’ evacuation targets the closest labeled constituent containing the root. However, Wiland (2026) proposes that Evacuate closest also cares about whether the closest labeled constituent contains ‘traces.’ Of course, Nanosyntactic trees do not contain traces, but the places where traces would have been can still be easily identified by single-branching nodes. In (64a), aP is a single-branching node within CMPRP.

The issue with this entry is that the derivation will never provide the right type of configuration for it to be inserted. To see this, recall first that when we have a positive degree where little *a* is expressed as a suffix, the structure is as in (66). When the positive corresponds to a single morpheme, the structure would be as in (67).



Both of these structures can give rise to a derivation where in the comparative, little *a* is expressed as a part of a suffix. In the case of (66), this leads to the disappearance of the stem marker. In the case of (67), this may give rise to suppletion. This has been discussed in detail in Sections 4.2-4.3.

However, it is impossible to merge CMPR to any of the structures in (66) or (67), and end up with a comparative structure where little *a* is expressed as a part of a *prefix*. Such a derivation would require backtracking to the stage where little *a* was merged as a feature, removing it, and merging it with an empty set in a second workspace. However, this is impossible given the ordering of the steps in the algorithm, repeated below from (45) for convenience.

- (68) The lexicalization algorithm (based on De Clercq et al. 2025, Wiland 2026)
- a. Merge F and lexicalize.
 - b. If a. fails, evacuate the closest labeled constituent that (a) contains the root and (b) has no single-branching nodes inside.
 - c. If b. fails, pied-pipe the immediately dominating node and lexicalize.
 - d. If c. fails, go back to the previous cycle, and try the next option for that cycle.
 - e. If d. fails, spawn a new derivation providing feature X and merge that with the current derivation, projecting feature X to the top node.

The point is that under backtracking, which is step (d), only the steps *preceding* backtracking are tried. Prefixation cannot be attempted because the algorithm hits the step (d) – more backtracking – before the step (e). Thus, the derivation cannot build a prefix under backtracking, as it backtracks even further before it does so. And then, after backtracking steps have been exhausted, the derivation returns to the stage where CMPR was merged as a feature, opens a new workspace and tries to build a Spec containing CMPR. However, the stage where the new workspace is opened corresponds precisely to the stage where – whatever the positive was – CMPR was first introduced. Thus, under the ordering in (68), the prefixation of CMPR always attaches the prefix on top of the form that corresponds to the positive: which is what we wanted to derive.³¹

³¹See Section 6 for an account of the suppletive quantifier *poveče* ‘more.’

Finally, it is worth pointing out that the very same reasoning holds for a potential scenario where the change-of-state verb would be derived by a prefix. Such a prefix would never be able to go down to little *a* and disturb the original lexicalization of the positive. This is only possible in a suffixing derivation like the one in (64). In other words, what we just derived, using the Nanosyntax tools, is the AAB conjecture in (69) (originally (9)).

- (69) The AAB conjecture
 Systematic AAB patterns arise in languages with analytic, pre-marking comparatives and suffixal change-of-state verbs.

4.5 Conclusions

This section showed that within Nanosyntax, it is possible to derive ABB patterns in suffixal languages and AAB in prefixal languages, using a specific formulation of Evacuate Closest proposed in Wiland (2026). We also showed that ABB derivations are impossible in prefixing languages if the formation of a Complex Left Branch follows all other steps of the algorithm, as in the original proposal by Starke (2018). What drives the distinction between prefixes and suffixes is that prefixes require the opening of a second derivational workspace, while suffixes only require a single workspace. According to Starke (2018), the opening of a second workspace is costly, and avoided at all costs, which is why this is the very last option in the algorithm. This, however, has the effect that prefixes never influence the shape of the form that has already been derived.

5 DM: locality domains

We turn now to a DM perspective on the distribution of the ABB and AAB patterns of stem markers. As already mentioned above, the key distinction between Czech and Ukrainian on the one hand, and Bulgarian and Turkish on the other, would most straightforwardly be treated in terms of locality, rather than linear order (pre- versus post-marking). Specifically, the RSG (40) mentioned above was treated in Bobaljik (2012) as a special case of a broader locality condition that requires the target and trigger of contextual allomorphy to be local to one another, where one locality domain is (to a first approximation) the morphological word (m-word). This will limit allomorphy of the stem marker (including deletion thereof) to contexts where the trigger is word-internal, just as it does for root allomorphy (suppletion). Since Bulgarian and Turkish have (generally) periphrastic comparatives, but synthetic deadjectival verbs, the comparative cannot govern irregularity (all patterns must be AAX), but the verbalizer (V_{Δ}) can: AAB. By contrast, Czech and Ukrainian may freely allow ABB patterns (as well as AAA). The main contrast thus reduces to a special case of patterns already well described in the literature on suppletion: the RSG. The following section presents the details of the account, and then turns to remaining open questions.

5.1 Domains for contextual allomorphy

In DM, as in some other frameworks, allomorphy is most commonly encoded via rules of exponence or vocabulary insertion (see Gouskova and Bobaljik 2024 for an overview). The rules in (70) describe the suppletive alternation for the Modern Greek adjective *kak-ós* ‘bad’ given in (71) (*-ós* is the masculine, nominative, singular ending.) Although the rules are written with a linear order, the inclusion of square brackets is intended to indicate that they make reference in principle to hierarchical structure rather than linear order. (70a) requires the adjectival root meaning ‘bad’ to spell out as *cheiró-* when it is in the local context of the comparative (the comparative marker itself has two allomorphs, suffixal *-ter*, and freestanding *pjo*).³²

- (70) a. BAD → cheiró / [__] CMPR]
 b. BAD → kak-

(71)

POSITIVE	COMPARATIVE
kak-ós	cheiró-ter-os * kak-(o)ter-os
kak-ós	pjo kak-ós * pjo cheiró-s

The Elsewhere Principle (Kiparsky 1973) ensures that the suppletive root will be used whenever its context is satisfied, thus synthetic *cheiró-ter-os* and not **kak(o)-ter-os*. However, in the simple periphrastic comparative, the suppletive root is not used: **pjo cheiró-s*.³³ Bobaljik (2012, 68-72) argues (as mentioned in section 3.4 above) that this is because of a general locality condition on contextual allomorphy whereby trigger and target must be in the same domain. This continues a long tradition of investigating the conditions on contextual allomorphy (see Carstairs 1987; Bobaljik 2012; Embick 2010 and Moskal and Smith 2016, among many others). The RSG in (40) is one manifestation of this locality condition.

Putting aside irrelevant details, assume that (72) represents the underlying syntax shared between both synthetic and analytic comparatives.³⁴ In an analytic comparative, the root and stem marker (by hypothesis, the categorizing head *a*) combine (by head movement or some other operation) but the comparative marker retains its syntactic (and thus morphological) independence (72a). Since the trigger for suppletion is not in the same complex word as the root, (70a) cannot apply, and the default root

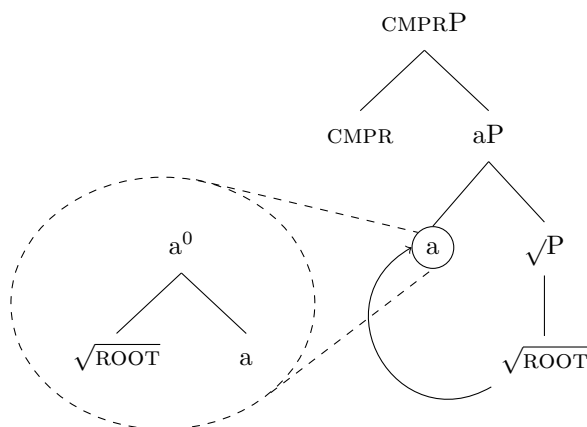
³²Following Bobaljik (2012) we omit the *a* head for ease of exposition. If it is included, one could posit that c-command, rather than structural adjacency, is the relevant condition for allomorphy: / [__] ... CMPR], or that *cheiro-* spells out the combination of the root and stem marker, or that the stem marker is ‘pruned’ (Embick 2010) (i.e., deleted) leaving the root and comparative structurally (and linearly) adjacent.

³³In Greek, as in a number of other languages, the picture is complicated by the pleonastic use of the pre-element ‘more’ as a reinforcer of a morphological comparative, thus doubly-marked *pjo cheiró-ter-os* is also possible. This doubling is independent of suppletion, cf. (varietal) English *more better*, *more stronger* etc.

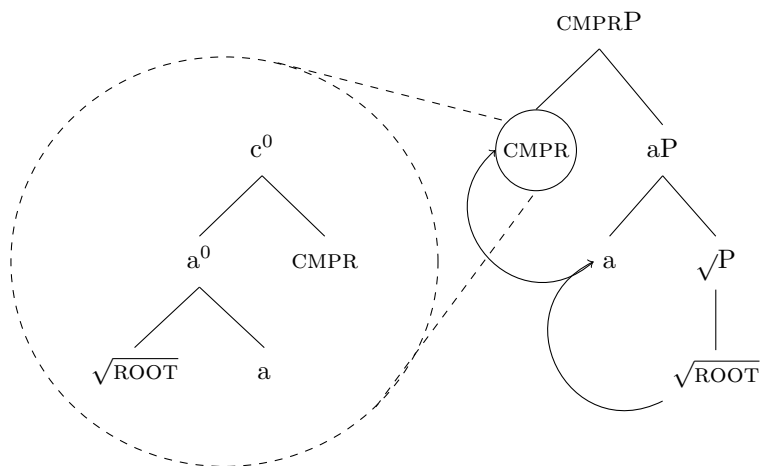
³⁴One way in which this is an oversimplification is in the position of inflectional morphology (agreement/concord in case, number and gender). In the synthetic construction, this morphology is peripheral to the comparative suffix, but in the periphrastic construction, the inflectional suffix occurs on the adjective, thus in some sense apparently closer to the adjective than the comparative marker. In recent work, Weisser (2026) argues that agreement morphology closes off the domain for contextual allomorphy in verbs, such that even when two morphemes are within the same m-word, allomorphic interactions across the locus of agreement are blocked. This might suggest an alternative characterization of the relevant locality domain than the one given here.

is inserted (70b): *pjo kak-ós*. In a synthetic comparative, the adjective and comparative head are combined together with the root (as in (72b)). Since the trigger for suppletion is in the same morphological word as the root, (71a) must apply, yielding *cheiró-ter-os*.³⁵

(72) a.



b.



5.2 Locality and ABB versus AAB

The considerations just discussed explain why Bulgarian and Turkish lack the ABB patterns seen in Czech and Ukrainian. Since the verbalizer, but not the comparative, forms a word together with the adjectival stem, only the verbalizer may trigger allomorphy on the root or on the stem marker: AAB is possible, unattested ABB is not. By contrast, in Czech and Ukrainian, the comparative is synthetic (suffixal), as in

³⁵We put aside here the question of whether (linear or structural) adjacency should be a part of the locality condition on contextual allomorphy. None of the examples we consider involve root suppletion across an overt but unchanging exponent of *a*; either null heads do not intervene or adjacency is not a condition on contextual allomorphy. Both views are espoused in the current literature. See note 32 and see Moskal and Smith (2016) for discussion.

(72c), thus CMPR may trigger root allomorphy and/or allomorphy of the stem marker, both of which are attested as we have shown.

A sample rule of exponence for the stem marker in Slavic languages might be as in (73). Here we simplify somewhat to make the general point, and elaborate further below.

$$(73) \quad a \rightarrow -(V)k \ / \]_{+K} \text{ --- }]$$

The rule in (73) inserts a *-k* stem marker after items that are diacritically marked to select for it.³⁶ We abstract away from whether the stem markers with and without a vowel preceding *k* are distinct affixes.³⁷

In Czech and Ukrainian, some roots are further specified to trigger the rule in (74). This can be read either as deletion of features (impoverishment in DM terms) or of the head (pruning, see Embick 2010) or insertion of a zero exponent; the choice is not relevant for the point being made here.³⁸

$$(74) \quad a \rightarrow \emptyset \ / \ \text{--- }] \text{ CMPR }]$$

If nothing further is said, then the assumption that the relevant change-of-state verbs embed the comparative (7) ensures that any stems to which (74) applies inherit that behaviour in the verb: ABB.

But in Bulgarian and Turkish, since the comparative is periphrastic in these languages, no rule of root or stem allomorphy that has the context $\text{--- }] \text{ CMPR }]$ (regardless of linear order) will be able to apply in the comparative. The comparative will always embed the positive unchanged: AAX.

By contrast, since the verbalizer is affixal, thus within the same morphological word as the stem marker, (75) is a possible rule.

$$(75) \quad a \rightarrow \emptyset \ / \ \text{--- }] \text{ ... } V_{\Delta}]$$

If there are no further contextual limitations on (75), then the pattern AAB is derived, which accords with the facts. Descriptively, the outcome of applying (75) with no morphological restrictions yields forms we could describe alternatively by saying the verbal suffix attaches (directly) to the root, rather than the stem, but recall that we take it to be the case that the semantics of the Bulgarian forms does not differ from Czech or Ukrainian, for which it has been argued that the verbs have comparative (degree achievement) semantics (see Bobaljik 2012, Chapter 6 for the general argument, and Vyshnevskaya 2025a for Ukrainian).

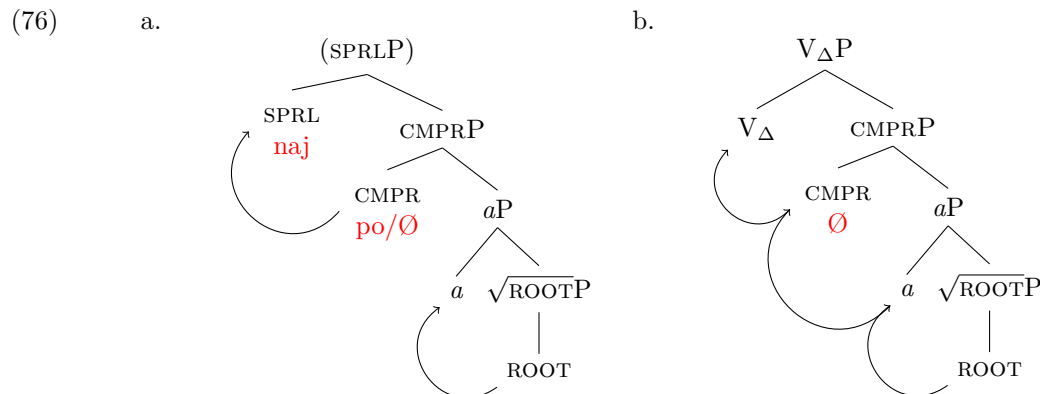
If the Bulgarian and Turkish change-of-state verbs were root-attaching, then describing the AAB patterns would be rather trivial, but also uninteresting in the

³⁶For roots, diacritic marking in DM corresponds roughly to root size in Nanosyntax – both are devices to regulate morphological combinatorial possibilities, often with no independently observable correlates. We will not enter here into a debate about the differences between the approaches on this point.

³⁷Recall from (11) and (26) that the diminutive affixes (*-ouč-k*, *-in-k*-, *-ič-k*, etc.) intervene between the root and the *-k* stem marker, and that the diminutives always include the stem marker, even if the root does not require it. In DM terms, this would be modeled by assigning the diacritic marking $[+K]$ to the diminutive exponents as well as to select roots. For arguments for the decomposition of the diminutive, along with a Nanosyntax analysis of these facts using morpheme size instead of diacritics, see Vanden Wyngaerd et al. (2025). For Turkish, Atlamaz and Dikmen (2026) do not further decompose the diminutive.

³⁸The difference may bear on the ways available to ensure that (74) bleeds (73).

sense that there would be no containment relationship between the verb and the comparative. Yet we have argued (for Bulgarian in particular) that the semantics of the verbs in Bulgarian are in relevant respects the same as for Czech and Ukrainian, that is, semantically (and therefore syntactically) embedding the comparative. That is, syntactically, we assume that the change-of-state verb, like the superlative, embeds the comparative, but that morphologically, the comparative head is realized as an independent m-word in the comparative (and superlative) but is incorporated into the verb and lacks a distinct realization in the verb, as in (76a) versus (76b). We include the superlative (though we do not discuss it here) for a more minimal contrast. The comparative has the exponent *po* (in red in the tree) when it is free-standing, but has zero exponence when it is contained in a more complex m-word (either the superlative or the change-of-state verb).³⁹



As a technical curiosity we note that in place of (75) these structures would allow us to posit the same rule in Bulgarian and Turkish as in Czech and Ukrainian, (74), and hold that in all the languages it is the comparative head that triggers zero exponence of the stem marker. But since the comparative is only in the same m-word as the root in the deadjectival verb, the rule can only apply in the verb. Observationally, these are equivalent, so we do not push the issue, but note in passing that this would allow the rule to apply locally.

5.3 Summary

In sum, existing assumptions within DM explain (with the caveat just mentioned) why Bulgarian and Turkish lack the ABB patterns of the stem marker that are frequent in Czech and Ukrainian. Conversely, the same assumptions allow for these patterns in Czech and Ukrainian which have suffixal comparatives and verbalizers, but do not require them. And of course, nothing in the theory requires that there be any allomorphy at all; any of the languages that we have considered could have had AAA patterns throughout; whether a language has allomorphy rules affecting stem-markers

³⁹ An alternative suggested in Bobaljik (2012) recapitulates the history of Bulgarian and posits a null comparative throughout, with *po* serving a reinforcing function (compare Greek and Italian as mentioned in note 21). See also note 41 below.

or not is a contingent fact. What the theory offered here restricts is merely the possible structural (or linear) relationships between target and trigger in rules such as (74) and (75).

6 Bulgarian redux: much-more ABA

We have noted above that Bulgarian adjectives differ from the rest of Slavic in having an intonationally independent pre-element rather than a suffix, as the sole overt morphological mark of the comparative grade (recall (37)), and in lacking suppletion in the comparative (23). Both the Nanosyntax and the DM accounts sketched here attempt to relate these two points of difference.

A problematic pattern that seems to make these results difficult for both accounts is the one attested in Bulgarian for the quantifier ‘many, much,’ depicted in (77), in which it appears that the root undergoes suppletion triggered by a pre-element which for the rest of the system appears to be free-standing.

(77)

GLOSS	POSITIVE	COMPARATIVE	SUPERLATIVE
‘many/much’	mnogo	po- veče	naj- mnogo

However, there are several differences between *poveče* ‘more’ and other comparatives that suggest that the *po-* in *poveče* is not prosodically and morphologically free in the way it is with all other adjectives. The first difference pertains to the stress pattern of *poveče*. Thus, recall from footnote 20 and the surrounding discussion that when an adjective such as *nóv* ‘new’ combines with *pó-*, we get a form where each vowel is stressed, i.e., *pó-nóv* ‘newer’ and similarly with all adjectives. The suppletive *póveče* differs from all comparatives and shows only a single stress, behaving in this respect as a single word.

The stress facts go hand in hand with the (im)possibility of coordination under a single comparative marker. This, recall, is generally possible with adjectives and illustrated in (78).

- (78) Tazi kniga beše **po-** [interesn-a i zabavn-a] od predišna-ta
 this piece was more interesting and entertaining from previous.DEF
 ‘This book was more interesting and entertaining than the previous one.’

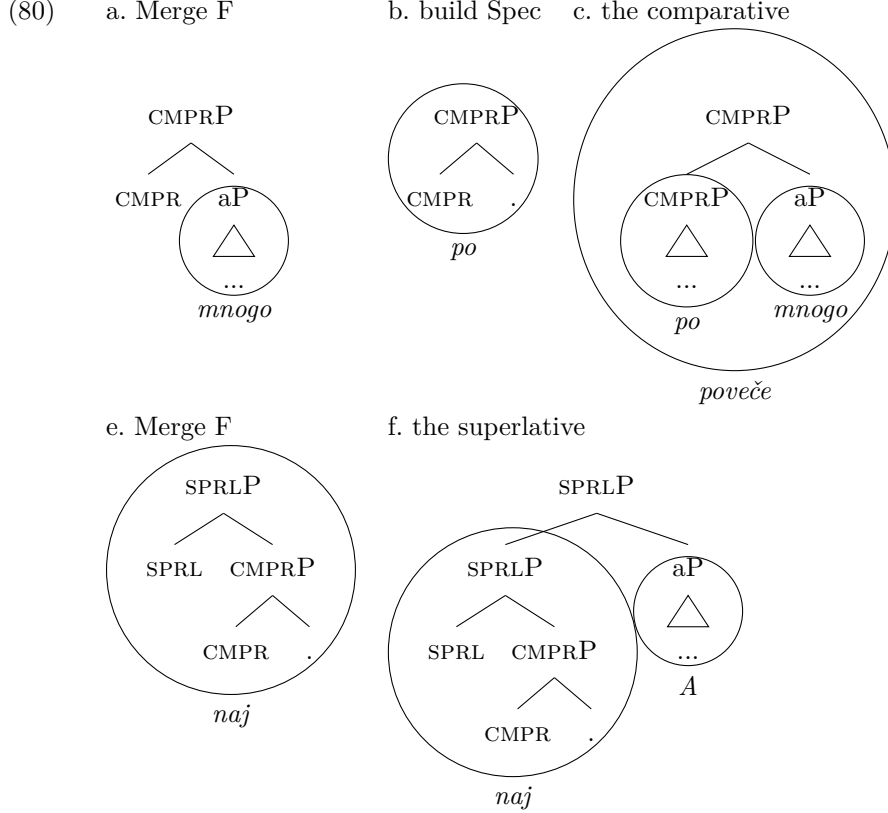
However, *poveče* ‘more’ cannot be used in the same way, regardless of the order of the conjuncts, see (79a,b). The relevant meaning can only be expressed when *po* is repeated in each conjunct, see (79c).

- (79) a. *Toj jadeše **po-** [veče i dølgo] od vsički ostanalite
 he ate CMPR much and long than all others
 b. *Toj jadeše **po-** [dølgo i veče] od vsički ostanalite
 he ate CMPR long and much than all others
 c. Toj jadeše [**po**-veče i **po**-dølgo] od vsički ostanalite
 he ate CMPR-much and CMPR-long than all others

‘He ate more and longer than everyone else.’

The question now is how these facts can be captured without undermining our overall accounts, both Nanosyntax and DM. In Nanosyntax, one conceivable analysis treats *poveče* as a non-decomposable form. Since a non-decomposable form has no prefix, the main generalization (banning suppletion under prefixation) is preserved.

It may appear that the simplest way of implementing this would be to say that the structure in (80a) is realized as *poveče* ‘more.’ However, such an analysis does not work because it predicts that the superlative would be based on the comparative form, i.e., on *poveče*. The reason is that under the system developed in Section 4, a prefix must be always added to an unchanged form. In this case, we would thus predict the superlative *naj-poveče*, an ABB pattern.

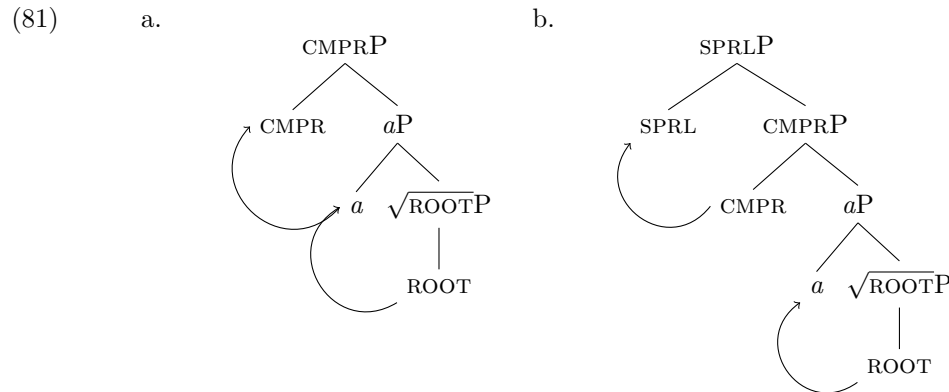


We thus assume that in deriving the comparative, the lexicalization algorithm forces the opening of a new workspace where the comparative marker is formed, see (80b). The item *poveče* applies only when the CMRP feature is added to *mnogo* as a complex left branch, see (80c).⁴⁰

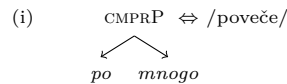
⁴⁰The tree in (80c) depicts the output of the lexicalization. We assume that the lexical item targeting this node contains pointers, as in (i). See Vanden Wyngaerd et al. (2021) for a discussion of pointers.

This analysis is compatible with the regular superlative *naj-mnogo* ‘most.’ The superlative is derived by merging the superlative feature into the auxiliary workspace as in (80d). This is a standard procedure where auxiliary workspaces (such as the one in (80b)) are only closed when lexicalisation inside them either fails or the derivation of the form terminates (see Starke 2018, Caha et al. 2019). Since there is no irregular item targeting the top node in (80e), the superlative is fully regular.

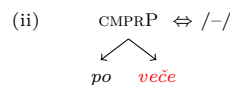
A more or less analogous account is available within DM. Of note here is that the comparative element forms an immediate constituent with the root in (80c) but in the superlative (80e) it forms an immediate constituent with the superlative. Bobaljik (2012, 86) proposes that (some) languages with affixal comparatives but periphrastic comparatives have derivations in which the comparative element CMPR combines with the adjectival root in the comparative (81a), but directly with the SPRL head in the superlative (81b).



Evidence for these derivations are languages which, like Bulgarian and Macedonian quantifiers, have suppletion in the (affixal) comparative but not in the (periphrastic) superlative. Bobaljik (2012, 77) provides examples from Tikhvin Karelian (82a) and Russian (82b) that fit this pattern.



Another line of analysis may invoke ‘private lexical entries’ in the sense of De Clercq and Vanden Wyngaerd (2019). Building on ideas by Starke, De Clercq and Vanden Wyngaerd (2019) propose that some lexical items (*veče*) are only accessible in a derivation in the context of other entries (*po-* in our case). They are used in a way that they only get activated when we have a lexical item as in (ii), with a pointer to such an entry (marked in red, as a diacritic for highlighting the fact that it is only accessible when pointed to).



We leave the choice between these two different implementations unresolved.

(82)

	GLOSS	POSITIVE	COMPARATIVE	SUPERLATIVE
a.	‘good’ (Tv.Kar)	hüvä	pare-mbi	ülen hüvä
b.	‘bad’ (Rus)	plox-oj	xuž-e	samyj plox-oj

Under such an account, the *po-* in *poveče* could be treated as the same *po-* as in all other comparatives. The difference in stress would be (presumably) a function of whether it is parsed as an affix (thus part of the same prosodic word as the root, and therefore within a single stress domain) or as a prosodically independent element (thus receiving its own stress).⁴¹

Ultimately, both Nano and DM accounts say something special about *poveče*. What we have just sketched shows that it can be done in either framework without running afoul of the (assumptions underlying) the RSG or GOSP which are doing the work in explaining the differences in the distribution of ABB versus AAB throughout this paper. Here too, though, there is a subtle difference between Nanosyntax and DM in the way the intuition about the special idiosyncrasy of *poveče* is cashed out. In Nanosyntax, idiosyncratic differences are primarily (or exclusively) encoded as differences in listed lexical entries: this requires that *poveče* be listed as such. In DM, there is also room for variation in the syntactic and post-syntactic operations that constrain vocabulary insertion. It is thus possible to posit that CMPR may be lexically specified to combine into a single m-word with some roots but not others while still treating it as a single exponent in both occurrences. But even here, how deep the difference lies between the approaches is not entirely clear: if the kind of head-building movement in (82) is syntactic, then the difference between *poveče* and a regular adjective such as *po-malək* ‘smaller’ (21) is also ultimately encoded in the lexical entries, either in terms of selecting features (probes) on CMPR or movement-driving features on the abstract morpheme $\sqrt{\text{MANY}}$, which has two listed allomorphs.

7 Conclusions

We started this paper with the question of whether AAB patterns exist in change-of-state verbs relative to the corresponding positive and comparative adjectives. For root suppletion, this was left as unresolved, and possibly unresolvable, in Bobaljik (2012). By turning to suppletive allomorphy (including zero allomorphy) of stem markers, we find a positive answer in Bulgarian and Turkish. This reaffirms the intuition in Bobaljik (2012) that *ABA and *AAB are not coextensive and should be treated distinctly. On the other hand, we found that AAB patterns are not freely available in this domain, even in our small sample: Czech deadjectival causatives and Ukrainian

⁴¹Bobaljik 2012, 131-132 suggested for this paradigm (and its cognate in Macedonian) a null comparative (suffixal) head, treating *po-* not as the head CMPR but instead as a reinforcer of the comparative meaning, as in the doubly marked comparatives mentioned in note 21. The palatalization on *č* in *poveče* is historically the remnant of the comparative suffix which Bulgarian shared with the rest of Slavic until about the fourteenth century (Reiter 1979, 21). If this null CMPR combines with the root in the comparative, but with the superlative marker in the superlative, then the apparent ABA pattern is readily described. This approach has parallels to Nanosyntax approaches which decompose the comparative into a series of comparative elements (Caha et al. 2019 et seq.). The account in the text is simpler in the sense of lacking an appeal to a null head, and is supported by the different prosodic status of *po-* in *poveče*.

productive inchoatives lack AAB patterns and have instead ABB patterns, which Bulgarian and Turkish lack. In the spirit of the workshop which spawned this paper, we approached this distribution jointly from the perspectives of Nanosyntax and DM: we showed that the two frameworks naturally meet in terms of relating the differing distribution of AAB and ABB patterns to independent differences among the languages: only Czech and Ukrainian in this group have suffixal comparatives. Bulgarian and Turkish comparative markers are prosodically independent elements that precede the adjective. We have shown that both frameworks readily derive the different distribution of patterns of stem-marker allomorphy from assumptions already established for patterns of root suppletion, in both cases holding that the two heads involved in an allomorphic alternation must be in the same domain, but the frameworks as currently conceived diverge in terms of the property that they take to be criterial for defining domains: for DM the relevant generalization is the RSG – the domain is (something like) the morphological word, where for Nanosyntax it is the special way in which prefixes are derived (GOSP). While the facts we consider here do not resolve the difference between the frameworks, we hope to have nevertheless contributed to the topic of this special issue by presenting in some detail one way in which we see a difference on which empirical evidence could in principle bear and thus an invitation for future work.

Acknowledgements. The authors wish to thank the organizers of the DM meets Nano conference as well as the audience for valuable comments on our presentation of this material.

Pavel Caha’s work was supported by the European Regional Development Fund project “A lifetime with language: the nature and ontogeny of linguistic communication (LangInLife)” (reg. no.: CZ.02.01.01/00/23_025/0008726).

References

- Atlamaz, Ü. and F. Dikmen. 2026. Derivational affixes as heads and phrases: evidence from turkish diminutives. *Morphology* 36(1): 2 .
- Blix, H. 2022. Interface legibility and nominal classification: A nanosyntactic account of kipsigis singulatives. *Glossa: a journal of general linguistics* 7(1) .
- Bobaljik, J. 2012. *Universals In Comparative Morphology*. Cambridge, MA: MIT Press.
- Bobaljik, J.D. and U. Sauerland. 2018. ABA and the combinatorics of morphological features. *Glossa: a journal of general linguistics* 3(1) .
- Borer, H. 2005. *Structuring Sense* (2 vols). Oxford: Oxford University Press.
- Caha, P. 2025. Root and stem allomorphy without multiple exponence: the case of special nominatives. *Morphology* 35(2): 97–141 .

- Caha, P. 2026a. Minimalism and Nanosyntax: Reconciling late insertion and the Borer–Chomsky conjecture, In The Cambridge Handbook of Minimalism and its Applications, eds. Leivada, E. and K. Grohmann, 57–82. Cambridge: Cambridge University Press.
- Caha, P. 2026b. Nanosyntax: Some key features, In The Cambridge Handbook of Distributed Morphology, eds. Alexiadou, A., R. Kramer, A. Marantz, and I. Oltra-Massuet, 31–73. Cambridge: Cambridge University Press.
- Caha, P., K. De Clercq, and G. Vanden Wyngaerd. 2019. The fine structure of the comparative. Studia Linguistica 73(3): 470–521. <https://doi.org/https://doi.org/10.1111/stul.12107> .
- Caha, P., K. De Clercq, and G. Vanden Wyngaerd. 2023. Zero morphology and change-of-state verbs. Zeitschrift für Sprachwissenschaft 42(1): 35–62 .
- Caha, P., K. De Clercq, and G. Vanden Wyngaerd. 2024. Allomorphy without context specification: a case study of czech adjectival stems. Morphology 34(4): 457–500 .
- Carstairs, A. 1987. Allomorphy in Inflexion. Croom Helm Linguistics Series. London: Croom Helm.
- De Belder, M. and J. Van Craenenbroeck. 2015. How to merge a root. Linguistic Inquiry 46(4): 625–655 .
- De Clercq, K., P. Caha, M. Starke, and G. Vanden Wyngaerd. 2025. Nanosyntax: State of the art and recent developments, In Nanosyntax and the Lexicalisation Algorithm, eds. Caha, P., K. De Clercq, and G. Vanden Wyngaerd, 1–44. Oxford: Oxford University Press.
- De Clercq, K. and G. Vanden Wyngaerd. 2018. Unmerging analytical comparatives. Jezikoslovlje 19(3): 341–363 .
- De Clercq, K. and G. Vanden Wyngaerd. 2019. On the idiomatic nature of unproductive morphology, In Linguistics in the Netherlands, eds. Berns, J. and E. Tribushinina, 99–114. Amsterdam: Benjamins.
- Dees, J. and K. VanDyne. in press. On the *ABA patterns of Polish inchoative deadjectival change-of-state verbs. Journal of Slavic Linguistics .
- Embick, D. 2010. Localism versus Globalism in Morphology and Phonology. Cambridge: MIT Press.
- Gouskova, M. and J.D. Bobaljik. 2024. Allomorphy and vocabulary insertion, In The Cambridge handbook of Distributed Morphology, eds. Alexiadou, A., R. Kramer, A. Marantz, and I. Oltra-Massuet. Cambridge: Cambridge University Press. In press.

- Kasenov, D. 2025. ABA in Russian adjectives, subextraction, and Nanosyntax, In Advances in Formal Slavic Linguistics 2022, eds. Gehrke, B., D. Lenertová, R. Meyer, D. Seres, L. Szucsich, and J. Zaleska, 255–292. Berlin: Language Science Press.
- Kennedy, C. and B. Levin. 2008. Measure of change: The adjectival core of degree achievements, In Adjectives and adverbs: Syntax, semantics and discourse, eds. McNally, L. and C. Kennedy, 156–182. Oxford: Oxford University Press.
- Kilgariff, A., V. Baisa, J. Bušta, M. Jakubíček, V. Kovář, J. Michelfeit, P. Rychlý, and V. Suchomel. 2014. The sketch engine: ten years on. Lexicography 1: 7–36 .
- Kiparsky, P. 1973. “elsewhere” in phonology, A Festschrift for Morris Halle, 93–106. New York: Holt, Rinehart and Winston.
- Leafgren, J. 2011. A Concise Bulgarian Grammar. SEELRC.
- Marantz, A. 1997. No escape from syntax: Don’t try morphological analysis in the privacy of your own lexicon, In Proceedings of the 21st Penn Linguistics Colloquium, eds. Dimitriadis, A., L. Siegel, C. Surek-Clark, and A. Williams, Volume 4, 201–225. Philadelphia: U Penn Working Papers in Linguistics.
- McFadden, T. 2018. *ABA in stem-allomorphy and the emptiness of the nominative. Glossa 3(1): 8 .
- Moskal, B. and P.W. Smith. 2016. Towards a theory without adjacency: Hyper-contextual vi-rules. Morphology 26: 295–312 .
- Reiter, N. 1979. Komparative. Wiesbaden: Harrassowitz.
- Scheer, T. 2003. The key to Czech vowel length: Templates, In Investigations into Formal Slavic Linguistics, eds. Kosta, P., J. Blaszczak, J. Frasek, L. Geist, and M. Zygis, 97–118. Frankfurt am Main: Peter Lang.
- Starke, M. 2009. Nanosyntax: A short primer to a new approach to language. Nordlyd 36: 1–6 .
- Starke, M. 2018. Complex left branches, spellout, and prefixes, In Exploring Nanosyntax, eds. Baunaz, L., K. De Clercq, L. Haegeman, and E. Lander, 239–249. Oxford: Oxford University Press.
- Starke, M. and L.I. Dalen. 2026. Sub-extracting out of the Icelandic *ABA problem. Glossa: a journal of general linguistics 11(1) .
- Svenonius, P. 2004. Slavic prefixes inside and outside vp. Nordlyd 32(2) .
- Tabak, . and P. Caha 2025. Decomposing deadjectival verbs in turkish. Talk at Nanodays II, Brussels, <https://www.crissp.be/wp-content/uploads/2025/11/>

tabak_caha_v2_nano2.pdf.

- Türk, U. 2020. Tackling the augmentative puzzle in turkish. In Proceedings of the Workshop on Turkic and Languages in Contact with Turkic, Volume 5, pp. 155–162.
- Ulan, R. 1972. Some features of basic comparative constructions. Working papers on language universals 9: 117–162 .
- Van Craenenbroeck, J. 2025. Blixemes, pointers, and *ABA. Ms. CRISSP, https://static1.squarespace.com/static/5217e223e4b090faa01f8f2d/t/690df196f5f3821d69c56cfa/1762521494897/paper_blixemes_aba_v2.pdf.
- Vanden Wyngaerd, G., P. Caha, and K. De Clercq. 2025. The rich internal structure of gradable adjectives, In Generative Perspectives on Degrees. The Semantics and Morphosyntax of Scalarity, eds. Vyshnevskaya, A. and E. Cavarani, 11–41. Berlin: De Gruyter Mouton.
- Vanden Wyngaerd, G., K. De Clercq, and P. Caha. 2021. Late insertion and root suppletion. Revista Virtual de Estudos da Linguagem - ReVEL 19(18): 81–123 .
- Vanden Wyngaerd, G., M. Starke, K. De Clercq, and P. Caha. 2020. How to be positive. Glossa. A Journal of General Linguistics 5(1): 1–34 .
- Vyshnevskaya, A. 2022. There is more to the morphology of -š/-iś: the comparative in ukrainian. Linguistics in the Netherlands 39(1): 263–278 .
- Vyshnevskaya, A. 2025a. Comparative morphology across categories. Ukrainian adjectives, adverbs, and deadjectival verbs. dissertation, KU Leuven, Leuven.
- Vyshnevskaya, A. 2025b. Comparing Slavic comparatives, In Nanosyntax and the Lexicalisation Algorithm, eds. Caha, P., K. De Clercq, and G. Vanden Wyngaerd, 47–88. Oxford: Oxford University Press.
- Weisser, P. 2026, May. An empirical argument against AGR-heads in the syntax from non-peripheral agreement markers. Paper presented at the Salzburg Conference on Comparative Syntax (SalCoS).
- Wiland, B. 2023. Polish comparative adjectives and adverbs. Studies in Polish Linguistics 18(3): 97–143 .
- Wiland, B. 2026. ‘pre’and ‘post’marking in nanosyntax: the case of a prepositional dative in a dialectal Polish. Poznan Studies in Contemporary Linguistics 62(1): 101–121 .
- Zompì, S. 2017. Case decomposition meets dependent-case theories. Master’s thesis, Pisa: Università di Pisa.